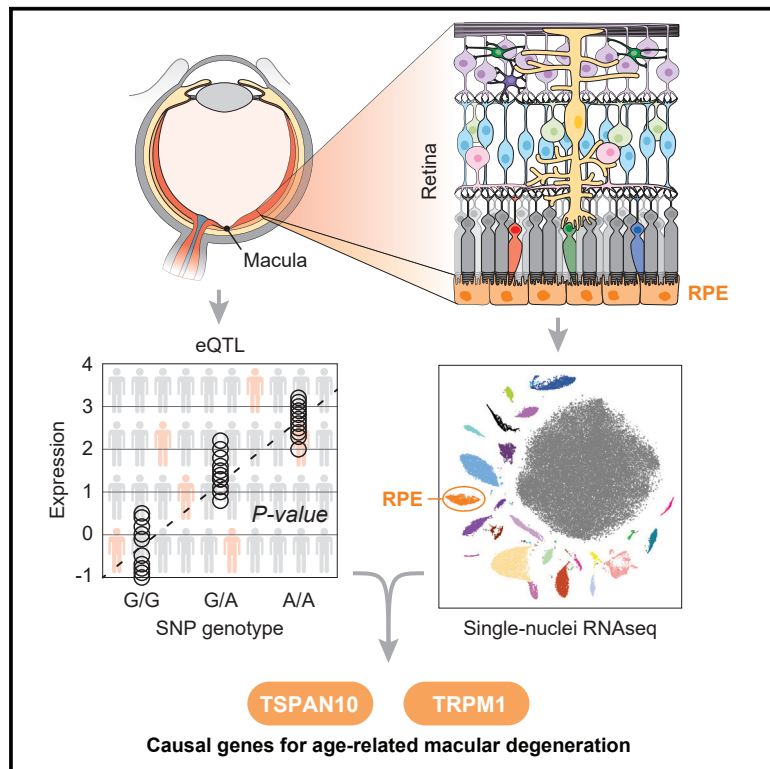


Cell Reports

Integration of eQTL and a Single-Cell Atlas in the Human Eye Identifies Causal Genes for Age-Related Macular Degeneration

Graphical Abstract



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In Brief

To find genes underlying risk for age-related macular degeneration, Orozco et al. analyze human ocular transcriptomes in conjunction with genotypes and build a single-nucleus atlas. They identify 15 putative causal genes, of which *TSPAN10* and *TRPM1* were enriched in retinal pigment epithelium, the site of early disease pathology.

Highlights

- A searchable database of eQTL from macula and peripheral retina and RPE/choroid
- Single-cell expression from human retina and RPE with cell subtype resolution
- Integrative analysis of bulk and single-cell transcriptomics in the human eye
- Co-localization of GWAS and eQTL identified putative causal genes for AMD

Data Resources

GSE135092
GSE135133



Integration of eQTL and a Single-Cell Atlas in the Human Eye Identifies Causal Genes for Age-Related Macular Degeneration

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SUMMARY

Age-related macular degeneration (AMD) is a leading cause of vision loss. To better understand disease pathogenesis and identify causal genes in GWAS loci for AMD risk, we present a comprehensive database of human retina and retinal pigment epithelium (RPE). Our database comprises macular and non-macular RNA sequencing (RNA-seq) profiles from 129 donors, a genome-wide expression quantitative trait loci (eQTL) dataset that includes macula-specific retina and RPE/choroid, and single-nucleus RNA-seq (NucSeq) from human retina and RPE with subtype resolution from more than 100,000 cells. Using NucSeq, we find enriched expression of AMD candidate genes in RPE cells. We identify 15 putative causal genes for AMD on the basis of co-localization of genetic association signals for AMD risk and eye eQTL, including the genes *TSPAN10* and *TRPM1*. These results demonstrate the value of our human eye database for elucidating genetic pathways and potential therapeutic targets for ocular diseases.

INTRODUCTION

Genetic and environmental factors underpin the onset and progression of common eye diseases such as age-related macular degeneration (AMD) (de Jong, 2006), glaucoma (Wiggs and Pasquale, 2017), and diabetic retinopathy (DR) (Wong et al., 2016). AMD affects 3 million people in the United States alone (American Macular Degeneration Foundation). It is characterized by the loss of photoreceptors and retinal pigment epithelium (RPE) in the macula, a cone-rich and primate-specific region in

the eye that is responsible for high-acuity vision. There are two recognized advanced forms of the disease, geographic atrophy (GA; or dry) AMD, and neovascular (wet) AMD. Therapies targeting vascular endothelial growth factor (VEGF) have revolutionized treatment for wet AMD (Rosenfeld et al., 2006), but not all patients respond to these medications (Yang et al., 2016), and there are currently no available treatments for dry AMD. The molecular mechanisms underlying susceptibility to macular degeneration are still unclear, and the absence of a macula in non-primate animal models such as the mouse highlights the need for comprehensive characterization of ocular tissues in humans.

AMD is a complex disease influenced largely by genetics. The most recent genome-wide association study (GWAS) for AMD (Fritsche et al., 2016) identified a total of 34 loci that account for 46% of the genetic variance. GWAS loci are a valuable resource for understanding disease mechanisms, but they do not necessarily pinpoint the causal genes responsible for the disease association. Genetic associations are often intergenic, and multiple studies have observed that causal genes are frequently located far from the peak GWAS association (Cannon and Mohlke, 2018; Gusev et al., 2016; Zhu et al., 2016). Expression quantitative trait loci (eQTL) can be used in conjunction with GWAS to facilitate the identification of candidate genes, and a recent study of eQTL in human retinal samples identified potential causal genes in 6 AMD GWAS loci (Ratnapriya et al., 2019). However, that analysis did not include RPE/choroid transcriptomic data, nor did it examine the differences between macula and peripheral tissues, both of which are highly relevant to AMD pathology.

Studies based on bulk RNA sequencing (RNA-seq) data from heterogeneous tissues are constrained by the lack of cell type resolution. In the retina, rod photoreceptors represent the majority of cells, and expression signals from cell types that are less abundant could be masked or diluted. Single-cell RNA-seq



(scRNA-seq) technology complements this deficiency but requires the researcher's immediate access to fresh tissue from surgical biopsy or autopsy. In recent years, single-nucleus RNA-seq (NucSeq) has enabled cell-type-specific gene expression analysis in complex tissues from frozen samples, especially in tissues of the central nervous system (Grindberg et al., 2013; Habib et al., 2017; Lake et al., 2017; Rosenberg et al., 2018; Sathymurthy et al., 2018).

To advance our understanding of ocular diseases, we present a rich human eye resource of macula and cell-type-specific expression profiles linked to genotypes. We performed (1) transcriptomics using bulk RNA-seq in RPE/choroid and retina tissues of both macular and non-macular regions (periphery) from 129 human donor eyes and genotyping of SNPs on 121 donor eyes, (2) eQTL analysis to identify *cis*-eQTL in human RPE/choroid and retina in bulk tissue samples from macula and non-macular regions, and (3) NucSeq in human eyes from five donors.

These datasets comprise human ocular transcriptional profiles and eQTL analysis with RPE/choroid representation, as well as NucSeq in human eyes with similar subtype resolution as previous studies on non-human primate and mouse (Macosko et al., 2015; Peng et al., 2019; Shekhar et al., 2016). We found that a majority of candidate genes for AMD were expressed in the RPE, and we used eQTL to identify 15 putative causal genes for 13 AMD risk loci. These included genes highly expressed in RPE, such as tetraspanin 10 (*TSPAN10*) and transient receptor potential cation channel subfamily M member 1 (*TRPM1*), as well as previously identified genes such as transmembrane protein 199 (*TMEM199*) and biogenesis of lysosome-related organelles complex 1 subunit 1 (*BLOC1S1*). Overall, our results advance our knowledge of the causal genes and cell types underlying AMD genetic risk and highlight how these datasets can be used by the broad ocular research community.

RESULTS

Large-Scale Profiling of Human Macular versus Non-macular RPE/Choroid and Retina

To meet the need for large-scale datasets in tissues relevant to AMD, we built a genome-wide expression RNA-seq database that included RPE/choroid and retina from macula and non-macular tissues, dissected from 129 postmortem donors (106 control and 23 AMD patients). Donor eyes were dissected for each tissue from the macula region and from the periphery away from the macula (non-macula; Figure 1A), resulting in a total of 529 human samples of four different tissue and region types (see STAR Methods). As expected, we observed distinct gene expression profiles in RPE/choroid and retina samples (Figures S1A–S1C; Data S1).

We found 754 genes differentially expressed (DE) between macula and non-macula regions in the retina and 968 genes DE between macula and non-macula RPE/choroid tissues (Figures S1D and S1E; Data S1). Genes DE between macula and non-macular regions largely overlapped with region-specific markers previously described in humans (Figures 1B and 1C) and in the macaque retina (Figure S1F; Li et al., 2014; Peng et al., 2019; Whitmore et al., 2014). Overall, we observed large

differences between macula or non-macula regions within the same tissue. Notably, multiple candidate genes for AMD, including *B3GLCT* (beta 3-glucosyltransferase), *HTRA1* (Htra serine peptidase 1), *CFI* (complement factor I), *TSPAN10*, *PILRA* (paired immunoglobulin like type 2 receptor alpha), and others, were different between macula and non-macular tissues (Data S1). In contrast, we observed very limited gene expression differences between AMD and controls in bulk RNA-seq datasets, possibly because of the relatively small number of AMD donors ($n = 23$) and heterogeneity in the complex eye tissues (Data S1).

NucSeq of Human Eyes

To characterize expression profiles of the many cell types of the human retina and RPE, we performed NucSeq in human eyes. From four post-mortem eyes, we profiled 100,055 nuclei after quality control (QC) and on average detected expression of 2,266 genes per nucleus (Figures S2A–S2C). To define cell types, we clustered the nuclei on the basis of their expression profiles and performed dimensionality reduction using t-distributed stochastic neighbor embedding (tSNE) to visualize the nuclei. We identified retinal cell types using known markers enriched in each cluster (Figures 2A and 2B; Data S2). Clustering was driven mainly by cell type and not by donor, batch, or sex (Figures S2D and S2E). All major groups were represented in our dataset (Figure 2A), including rod and cone photoreceptors, bipolar cells (BCs), amacrine cells (ACs), horizontal cells (HCs), retinal ganglion cells (RGCs), Müller glia, astrocytes, RPE, vascular cells, and myeloid cells. We found small but detectable differences in the proportions of cell types observed across the different donors (adjusted $p < 0.006$, Fisher's exact test), which were comparable with the level of variation observed across different replicates from the same donor (Figure S2D; Data S2). We also performed a pseudo-bulk analysis for each major cell type, in which we computed the average expression of each gene in a given cell type across all cells from one donor and performed differential expression at the donor level, rather than at the single-nucleus level. This approach ensures that the p values from the analysis are not inflated by the large number of nuclei and correctly normalizes for sequencing depth (Lun and Marioni, 2017; Shulze et al., 2019). Cell type markers identified using the Wilcoxon rank-sum test using Seurat (Data S2) were consistent with genes DE in each cell type using the pseudo-bulk approach (Data S3; Figure S3).

To further characterize retinal cell subtypes, we first excluded the rods from subsequent analyses (85,897; 70% of nuclei), which also removed the majority of cells containing high levels of ambient RNA that escaped QC filters. We observed a larger number of clearly defined clusters from the remaining cell types, including additional subtypes of bipolar and ACs. Using a similar approach, we re-clustered the nuclei from each main cell type: ACs, BCs, HCs, cone photoreceptors, RGCs, and non-neuronal cells (Figures 2C–2H). In 16,540 BC nuclei, we resolved 13 distinct clusters, broadly categorized into OFF and ON BCs using established functional marker genes (Data S2) such as *GRM6* (glutamate metabotropic receptor 6) and *GRIK1* (glutamate ionotropic receptor kainate type subunit 1; Figure 3A). On the basis of markers identified in previous mouse and primate retina single-cell studies (Macosko et al., 2015; Peng et al.,

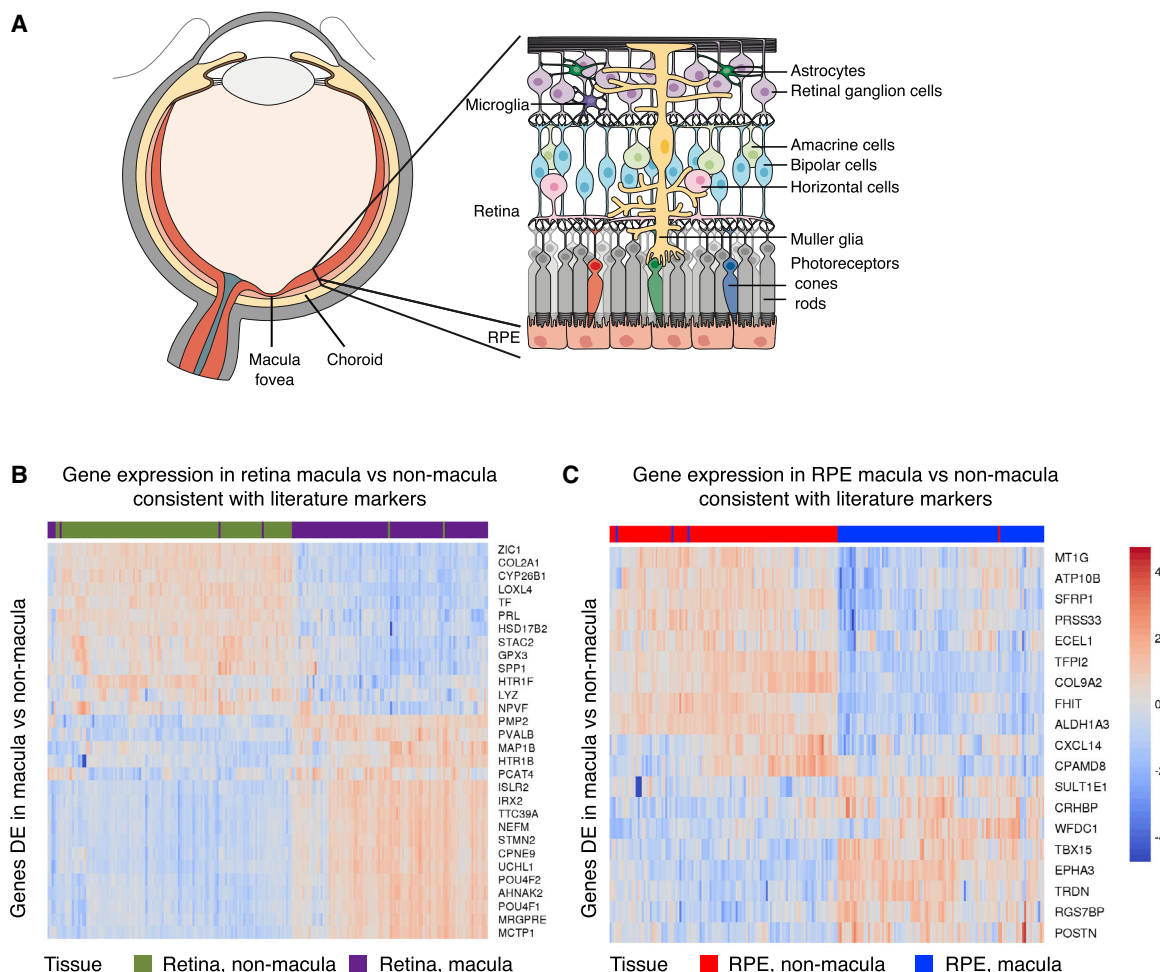


Figure 1. Macula and Tissue-Specific Transcriptomes from Human Retina and RPE

(A) Schematic of eye tissue cell types collected from the retina and RPE.

(B and C) Genes differentially expressed between macula and non-macula in our bulk RNA-seq samples were consistent with the literature. Genes shown in rows are DE in present and published datasets. Columns are samples, and genes come from studies (Peng et al., 2019; Voigt et al., 2019; Whitmore et al., 2014) for differential expression in the (B) retina (n = 124 donors) and (C) RPE/choroid (n = 125 donors).

See also Data S1 and Figure S1.

2019; Shekhar et al., 2016), we unambiguously assigned 12 known functional bipolar subtypes (Figures 3A–3C), including the foveal-specific OFFx cluster recently identified in the macaque retina. In addition, we identified a bipolar cluster that is molecularly similar to DB4 (Figure 3C), which we observed in all donor eyes. Top marker genes differentiating the 13 bipolar clusters are shown in Figure 3B.

In 6,101 AC nuclei, we identified multiple GABAergic and glycinergic amacrine clusters on the basis of the expression of *GAD1* (glutamate decarboxylase 1), *GAD2* (glutamate decarboxylase 2), and the glycine transporter *SLC6A9* (solute carrier family 6 member 9; Figure 3D). We found distinct clusters expressing functional and/or molecular marker genes for All, starburst, VIP, VGLUT3, and SEG amacrine subtypes, indicating the biological relevance of the sub-clusters. The 9 glycinergic amacrine clusters also correspond to the macaque AC glycinergic subtypes. We resolved 9 GABAergic subtypes in our dataset, fewer than

the 26 reported in the macaque study, and as a consequence it was challenging to match up the human and macaque subtypes. This difference in resolution was likely due to the smaller number of amacrine nuclei collected in our dataset. Nevertheless, some of the GABAergic clusters appear to contain multiple smaller clusters, which could potentially be further resolved with deeper NucSeq data. Top marker genes differentiating the 18 amacrine subtypes are shown in Figures 3E and 3F.

We collected 1,757 nuclei from RGCs and resolved 5 subtypes (Figure 2E). Four of these corresponded to the main subtypes also identified in macaques (Peng et al., 2019): OFF midgate (cluster 1, OFF MGC), ON midgate (cluster 2, ON MGC), OFF parasol (cluster 4, ON PGC), and ON parasol (cluster 5, ON PGC). Cluster 3 (RGC) may consist of rare subtypes that do not have enough nuclei to be resolved. Our results are in contrast to the 46 RGC subtypes identified in mouse retina at postnatal day (P) 5 (Rheume et al., 2018). Additional work examining a

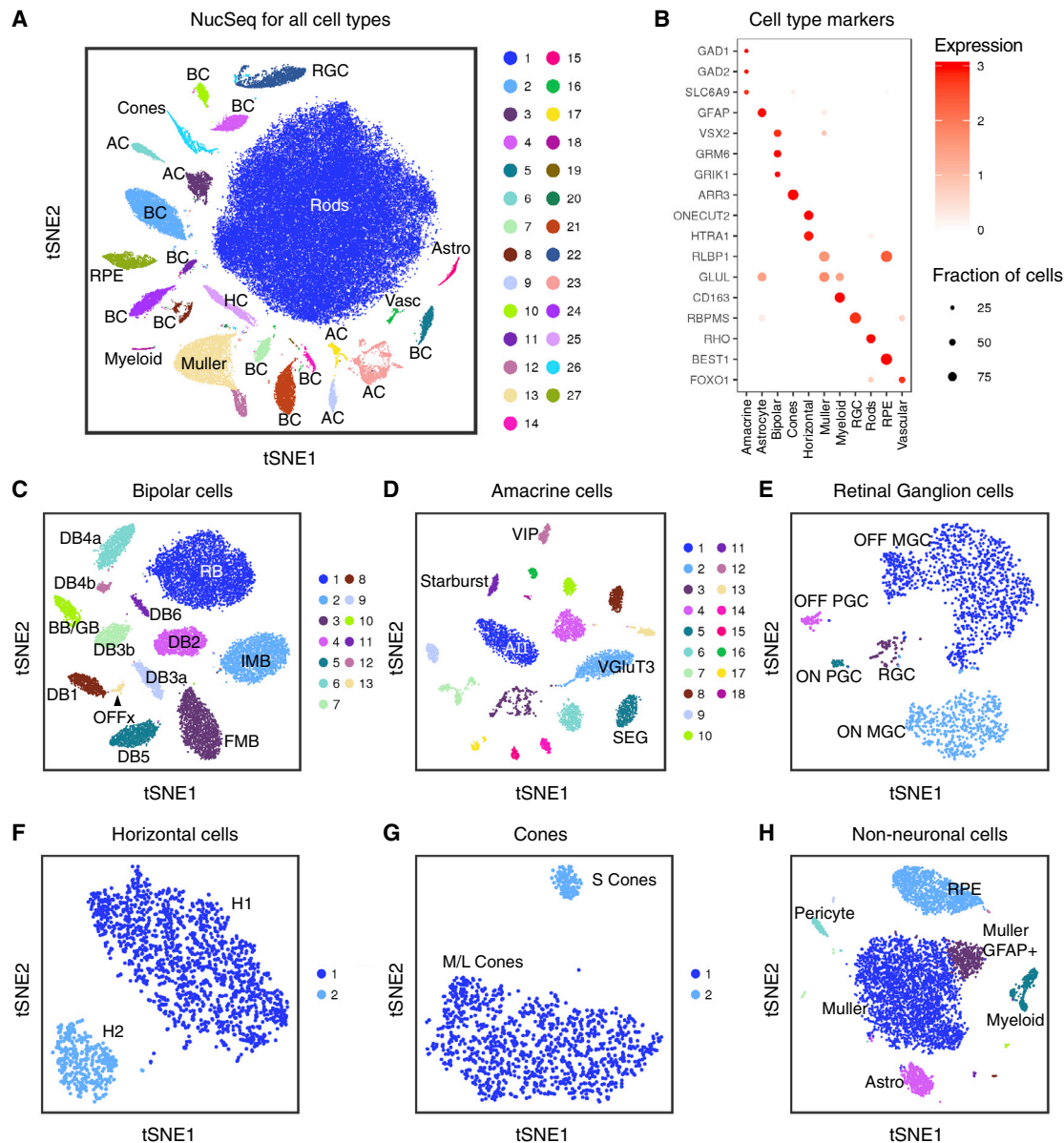


Figure 2. Single-Nucleus RNA Sequencing in Human Retina and RPE

(A and C–H) Dimensionality reduction using tSNE of NucSeq. Each dot is a nucleus. Nuclei are colored by cluster. Cell types shown are (A) all cell types observed, (C) bipolar, (D) amacrine, (E) retinal ganglion, (F) horizontal, (G) cone photoreceptors, and (H) all non-neuronal cells. Cluster labels are as follows: (A) rods are 1; bipolar cells (BC) are 2, 4, 5, 7, 8, 10, 11, 14, 19, 21 and 24; amacrine cells (AC) are 3, 6, 9, 17, and 23; Müller glia are 12 and 13; astrocytes are 15; non-neuronal cell types are 16 and 20; myeloid cells are 18; retinal ganglion cells (RGCs) are 22; horizontal cells (HC) are 25; cones are 26; and RPE is 27. Cluster labels in (D): numbers denote AC clusters 1–18, All AC is 1, vGluT3 AC is 2, SEG AC is 5, starburst AC is 11, and VIP AC is 12. Cluster labels in (H): Müller glia are 1 and 3, RPE are 2 and 12, astrocytes (Astro) are 4, myeloid cells are 5, pericytes are 6, melanocytes are 7, T lymphocytes are 8, endothelial cells are 9, Schwann cells are 10, and B lymphocytes are 11.

(B) Dot plot of top marker genes expressed in major cell types. Cell types are plotted on the x axis, genes are on the y axis, color of dots is the average expression of each gene in each cluster, and the size of the dot is the fraction of nuclei in the cluster expressing that gene.

For all panels, donor $n = 4$. See also [Data S2](#) and [Figure S2](#).

larger number of RGCs is needed to determine if there are additional subtypes of RGCs in adult human retina, or if this discrepancy is due to developmental or species-specific differences. Consistent with single-cell RNA-seq from macaque retina, we also observed two subtypes of HCs H1 and H2 ([Figure 2F](#))

and two clusters corresponding to cone photoreceptors M/L (cluster 1) and S cones (cluster 2; [Figure 2G](#)).

Within the non-neuronal cells, we collected 1,635 RPE nuclei as well as distinct cell types of Müller glia (5,024), astrocytes (508), myeloid cells (395), vascular cells including pericytes and

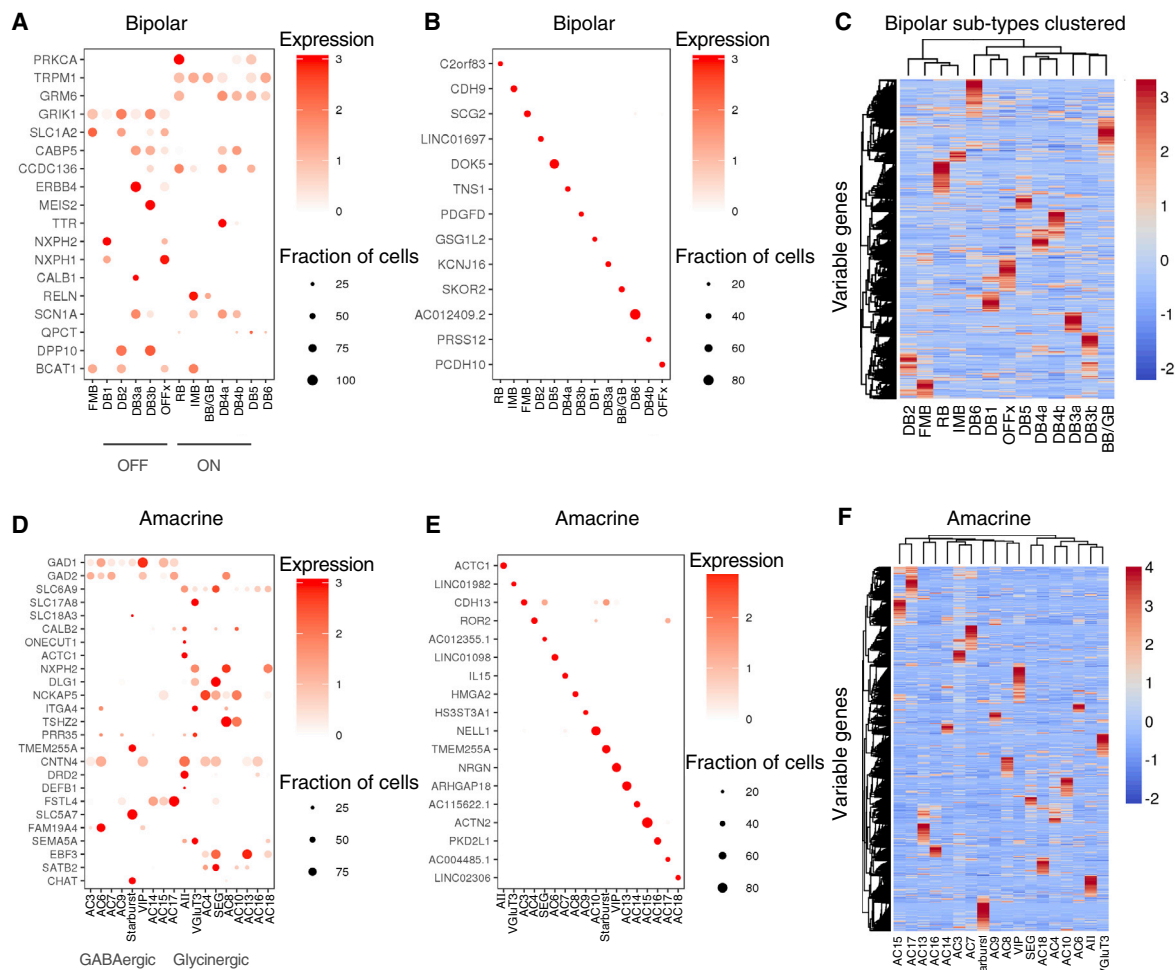


Figure 3. Cell-Type-Specific Expression in Retina and RPE/Choroid Using NucSeq

(A and B) Dot plot of marker genes for bipolar cell clusters (A) from the literature and (B) identified using Seurat. Rows are genes, columns are nuclei grouped by cluster, and the color is the expression level of each gene in each cluster. Cluster order in (B) is in descending cluster size.

(C) Clustering of bipolar clusters on the basis of expression of top 2,000 variable genes.

(D and E) Dot plot of marker genes for amacrine cell clusters (D) from the literature and (E) identified using Seurat. Cluster order in (E) is in descending cluster size.

(F) Clustering of amacrine clusters on the basis of expression of top 2,000 variable genes.

For all panels, donor $n = 4$. See also [Data S3](#) and [Figure S3](#).

endothelium (317), and small numbers of T cells, B cells, Schwann cells, and melanocytes, which likely originated in the choroid (Figure 2H). We identified a distinct Müller glia cluster that could not be merged with the larger Müller cluster, regardless of the clustering resolution we used. This Müller cluster showed increased expression of the glial activation marker GFAP (glial fibrillary acidic protein) and was specific to one donor eye. The gene signature of this GFAP+ population is consistent with activated Müller cells in hyperglycemic conditions (Coughlin et al., 2017), although there was no diabetes diagnosis in the limited clinical history of this donor.

Finally, NucSeq was able to detect a larger number of long noncoding RNAs (lncRNAs) relative to traditional bulk RNA-seq. Many of these lncRNAs were cell-type-specific markers, and hence, NucSeq would be a useful tool for studying lncRNAs in primate- and/or human-specific molecular mechanisms in the

eye. Our data showed the validity of using NucSeq from frozen eyes as a substitute for scRNA-seq of freshly dissected retina. The loss of cytosolic mRNAs did not prevent the segregation of cell subtypes, as the limitation of subtype resolution appears to lie in the number of nuclei for each cell type instead of the method itself. Overall, we present a comprehensive NucSeq dataset with subtype resolution from human retina, RPE, and choroid.

Cell-Type-Specific Expression Patterns of Retinal Disease-Related Genes

Candidate genes associated with AMD risk (Fritsche et al., 2016; Persad et al., 2017) provide a useful starting point for understanding the pathways and cell types involved in the disease. We found that non-neuronal cells expressed the largest number of AMD GWAS candidates, with ~38% (18) of genes expressed

in RPE cells, 29% (14 genes) in Müller glia, and 27% (13 genes) in astrocytes. A smaller number of genes were expressed in neuronal cell types (Figure 4A). *TSPAN10*, an AMD GWAS candidate gene of unknown function, was most highly expressed in RPE cells but was low or undetectable in every other cell type observed. Some candidates, such as *APOE* (apolipoprotein E), are known to be expressed in multiple tissues but were expressed primarily in Müller cells in the context of the retina. Our results are consistent with our current understanding of AMD, in which RPE, glia, and myeloid cells are thought to play important roles in the disease etiology.

We examined the expression profiles of genes harboring causal mutations for retinitis pigmentosa (RP; Farrar et al., 2017), congenital stationary night blindness (CSNB; Zeitz et al., 2015), neuronal ceroid lipofuscinosis (NCL; Warrior et al., 2013), and familial normal-tension glaucoma (NTG; Wiggs and Pasquale, 2017). We also examined candidate genes in GWAS loci for primary open angle glaucoma (POAG; Wiggs and Pasquale, 2017) and diabetic retinopathy (DR; Graham et al., 2018; Grassi et al., 2011; Meng et al., 2018; Wang et al., 2014). As expected, the majority of causal genes for RP and CSNB were specifically expressed in photoreceptors and predominantly in the rods (Figures 4B and 4C). Genes highly expressed in the RPE were the next most abundant for both hereditary retinal diseases. Although visual cycle genes such as *RPE65* (retinoid isomerase), *LRAT* (lecithin retinol acyltransferase), and *RLBP1* (retinaldehyde binding protein 1) have been thoroughly studied, several of the disease causal genes expressed in the RPE remain unclear. *Clarin 1* (*CLRN1*) is known to be mutated in Usher syndrome type III but with unclear disease mechanisms in the retina (Adato et al., 2002). Here we found that *CLRN1* was primarily expressed in Müller cells, suggesting that there is retinal glial involvement in this disease. In summary, our eye NucSeq dataset can be used to identify the cell types relevant to diseases biology for multiple ocular diseases that involve the retina or RPE/choroid (Figure 4D).

Human Eye eQTL Database for Retina and RPE/Choroid

To improve our understanding of how genetics influences gene expression in the eye tissues relevant for AMD, we turned to eQTL. We genotyped the majority of donor eyes from the aforementioned donors and used expression from the bulk RNA-seq dataset described above (see STAR Methods). We focused our analysis on local or *cis*-eQTL, which represent genetic variants influencing expression of nearby genes, because of the limited power provided by our cohort size for *trans*-eQTL analysis.

To find eQTL, we performed linear regression between gene expression and SNP genotypes for each gene and identified associations in macular and non-macular RPE/choroid and retina at a false discovery rate (FDR) less than 5% (see STAR Methods). Because nearby SNPs can be tightly linked during meiosis, we used pruning based on correlation, also known as linkage disequilibrium (LD) between nearby SNPs (LD pruning), and found 75,666 independent associations, corresponding to 18,935 genes with eQTL (eGenes) in at least one of the four eye tissues (Figure S4A). Half of all associated SNPs (50%) were found within 93 kb of the gene start, and 90% of associations were within 629 kb of the gene start (Figure S4B). Roughly

32% of genes (6,178) were eQTL in both the RPE/choroid and retina, but 3,810 genes (20%) were eQTL in the RPE/choroid and not the retina, and 8,947 genes (47%) were eQTL in the retina and not the RPE/choroid. Similarly, a large fraction were eGenes only in RPE macular tissues (30%) compared with RPE non-macula, or in retina macular tissues (27%) and not in retina non-macula tissues (Figures S4C and S4D). This result remained true when we used a more stringent significance threshold to call eQTL (FDR < 0.1%), although the fraction of tissue specific eGenes decreased to 22% for RPE-macula and 20% for retina-macula. Genes with eQTL in macula but not in peripheral tissues were expressed uniformly across the different cell types in our NucSeq data (Figure S4E), consistent with our finding that the majority of genes expressed in retina and RPE/choroid did not differ between macula and non-macular tissues.

To determine if gene expression was indeed regulated in *cis*, we performed a *cis-trans* test. In individuals heterozygous for coding SNPs, we can use SNP variants directly observed in exonic reads from RNA-seq data, to separate transcript read counts derived from each chromosome. If a gene is regulated in *trans*, we expect to see roughly an equal fraction of transcript coming from each chromosome, because *trans*-acting factors, for example transcription factors or repressors, act equally on both chromosomes. In contrast, if a gene is regulated in *cis* (i.e., by genetic variation in the same chromosome), we expect to see a larger fraction of the transcript coming from one of the two chromosomes. This is also known as allele-specific expression (see STAR Methods). Overall, we found a significant overlap between genes predicted to be regulated in *cis* on the basis of allele-specific expression and *cis*-eQTL in all four eye tissues examined (Figure S4F).

Eye eQTL Identifies Causal Genes for AMD GWAS Loci

Although transcriptomic analyses have provided insight into the genes and pathways expressed in each cell type and tissue, they did not define the causal genes underlying disease risk susceptibility. Previous GWAS have identified a total of 35 loci (Fritsche et al., 2016; Persad et al., 2017) associated with risk of AMD, corresponding to dozens of candidate genes. Therefore, we used our eye eQTL database to prioritize candidate genes underlying GWAS loci for risk of AMD. To determine if a gene is causal for a disease association, we used co-localization of the GWAS and eQTL genetic signals in each locus (Giambartolomei et al., 2014). In co-localization, instead of relying on association results at a single peak GWAS SNP, we look for association in both GWAS and eQTL across all SNPs in the locus. Results of the CO-LOC test range from 0 to 1, where COLOC results closer to 1 represent a higher likelihood that genetic variation at the locus is driving both eQTL and GWAS. Overall, we found 15 putative causal genes for AMD, where the same SNPs associated with risk of AMD are also driving expression of these genes. These included 5 genes for which the putative causal gene on the basis of COLOC is different than the candidate gene originally defined on the basis of proximity (Table 1; Data S4).

As an example, a region in chromosome 17 has been associated with increased risk of AMD. When we explored the region, we found multiple candidate genes physically located within a block of LD surrounding the associated SNP, including

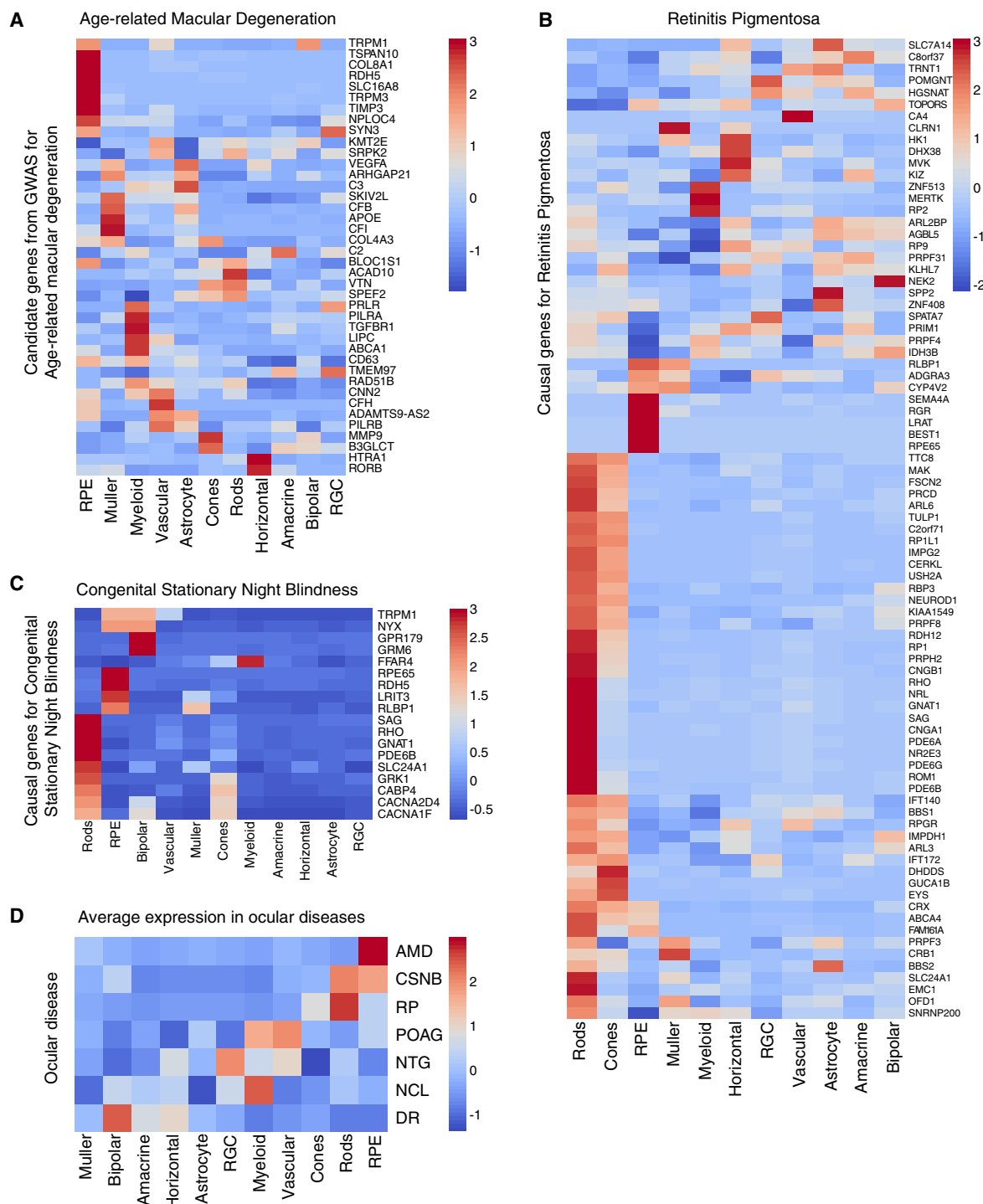


Figure 4. Cell-Type-Specific Expression in Ocular Disease Susceptibility Genes

(A–C) Rows are genes, columns are cell types identified in NucSeq data, and the color is the average expression of a gene in a given cell type. (A) Candidate genes from the most recent GWAS for risk of AMD. (B) Causal genes known for retinitis pigmentosa. (C) Causal genes known for congenital stationary night blindness. (D) Known or candidate genes for multiple ocular diseases. Ocular diseases are shown in rows, columns are cell types identified in NucSeq data, and the color is the average expression for all genes associated with a given disease, for each cell type. AMD, age-related macular degeneration; CSNB, congenital stationary night blindness; RP, retinitis pigmentosa; POAG, primary open angle glaucoma; NTG, normal-tension glaucoma; NCL, neuronal ceroid lipofuscinosis; DR, diabetic retinopathy.

For all panels, donor $n = 4$.

Table 1. Causal Genes in GWAS Loci for Risk of AMD

Locus Name	Putative Causal Gene	eQTL SNP	eQTL p Value	eQTL Tissue	Distance to GWAS (bp)
PILRB-PILRA	<i>PILRB</i>	rs6965458	5.72E-29	retina, macula	–18,146
PILRB-PILRA	<i>PILRA</i>	rs28495773	2.26E-23	retina, macula	–47,965
SLC16A8	<i>BAIAP2L2</i>	rs760975	2.48E-11	RPE, non-macula	–8,492
NPLOC4-TSPAN10	<i>TSPAN10</i>	rs7405790	9.06E-10	RPE, non-macula	94,809
B3GALT	<i>B3GLCT</i>	rs4381465	1.94E-08	retina, macula	–113,765
TRPM1	<i>TRPM1</i>	rs12440880	3.05E-08	RPE, non-macula	–21,976
MMP9	<i>SLC12A5-AS1</i>	rs3761159	2.85E-07	retina, non-macula	19,966
RDH5-CD63	<i>BLOC1S1</i>	rs56108400	4.78E-07	retina, non-macula	97,519
RDH5-CD63	<i>RDH5</i>	rs3138142	3.33E-05	RPE, non-macula	–193
TMEM97-VTN	<i>TMEM199</i>	rs7212510	5.07E-07	retina, macula	53,965
CTRB2-CTRB1	<i>BCAR1</i>	rs9928736	3.83E-05	retina, macula	7,857
COL4A3	<i>COL4A3</i>	rs7559693	5.47E-05	retina, non-macula	32,029
TNFRSF10A	<i>TNFRSF10A</i>	rs11777697	7.03E-05	RPE, non-macula	6,013
ARMS2-HTRA1	<i>HTRA1</i>	rs11528744	1.66E-04	retina, macula	78,089
CFI	<i>CFI</i>	rs141098594	2.18E-04	RPE, macula	–145,699

See also [Data S4](#) and [Figure S4](#).

TSPAN10 and *NPLOC4* (NPL4 homolog, ubiquitin recognition factor), both of which are expressed in the eye. *TSPAN10* is a *cis*-eQTL in RPE/choroid but not in the retina, but we found no eQTL for the neighboring gene *NPLOC4* (<http://eye-eqtl.com/>). eQTL analysis showed that people carrying the risk SNP for AMD in this locus (rs6565597) have decreased expression of *TSPAN10* (Figures 5A and 5B). Consistent with *cis* regulation on the *cis-trans* test, a greater fraction of *TSPAN10* transcript counts come from the strand with a C allele at coding SNP rs7210026, as well as additional coding SNPs in the gene (Figure 5C; Figure S5E). Co-localization of *TSPAN10* eQTL and AMD GWAS genetic signals (Figures 5D and 5E) suggested that genetic variation driving decreased expression of *TSPAN10* is also responsible for an increase in risk of AMD (COLOC probability = 0.96). Finally, we found that *TSPAN10* was primarily expressed in RPE cells in our eye NucSeq data (Figure 5F) and confirmed this expression pattern using *in situ* hybridization (Figure 5G). *TSPAN10* is an integral membrane protein in the TspanC8 family of tetraspanins, and expression of the gene appears to be restricted to melanin-producing cells such as RPE, melanocytes, and some brain tissues. Overall, this eQTL and expression pattern supports *TSPAN10* as the most likely causal gene in this locus.

We note that eQTL can be used not only to identify causal genes but can also be used to rule out candidate genes in a locus. Solute carrier family 13 member 2 (*SLC13A2*) is located near the *TMEM97-VTN* GWAS locus for AMD risk in chromosome 17. High expression levels and an eQTL for *SLC13A2* in the retina tissues suggested that this gene was a plausible candidate underlying risk of AMD (Figures S5A and S5B). However, the eQTL does not co-localize with the genetic association for AMD risk, de-prioritizing *SLC13A2* as a causal gene for this locus (COLOC probability = 0.17; Figures S5C and S5D). In contrast, *TMEM199* had an eQTL in retina macula (rs7212510, $p = 5.1 \times 10^{-7}$) and RPE macula tissues (rs241780, $p = 7.5 \times 10^{-5}$), and co-localization with AMD GWAS (COLOC probability = 0.98)

suggested that *TMEM199*, and not *SLC13A2*, was a likely causal gene for the *TMEM97-VTN* locus.

Finally, we demonstrate how these datasets can integrate genetics and genomics data to refine our understanding of genetic loci. GWAS for AMD risk has identified a nominal association near *TRPM1* on chromosome 15 (Fritsche et al., 2016) that was later replicated at the genome-wide significance threshold (Persad et al., 2017). In our database, we found a *cis*-eQTL for *TRPM1* in RPE/choroid tissue, but not in the retina (Figures 6A and 6B). *TRPM1* is regulated in *cis* on the basis of allele-specific expression, which is based on data from coding SNPs in *TRPM1* (Figure 6C). Co-localization of the genetic signal for association with AMD risk with the eQTL for *TRPM1* suggested that the same SNPs that drive increased risk also drive increased expression of the gene (COLOC probability = 0.98) and hence that *TRPM1* is likely the causal gene for this locus (Figures 6D and 6E).

TRPM1 is expressed in both the RPE and in retinal BCs in our NucSeq data (Figure 6H). We confirmed this expression pattern using *in situ* hybridization, in which we found expression of *TRPM1* in the RPE layer and in the inner nuclear layer, where BCs are located (Figure 6I). Although BCs are present in the retina, we did not detect an eQTL for *TRPM1* in this tissue (Figure 6B). Using our bulk RNA-seq data, we found a strong bias for differential start site use between retinal tissue and RPE/choroid (Figures 6F and 6G). The start site preferentially used in RPE/choroid tissue coincides with both the eQTL signal and the peak association for risk of AMD at the *TRPM1* locus. Taken together, these results suggest *TRPM1* expression in the RPE is responsible for the genetic risk of AMD, rather than its expression in BCs.

Our resource database comprises eQTL from both human RPE/choroid and retina that allow closer interrogation of disease-associated genes. We make our eQTL results available to the research community through an easy-to-use web interface: users can look for eQTL in our ocular tissues by searching for genes or SNPs at <http://eye-eqtl.com/>.

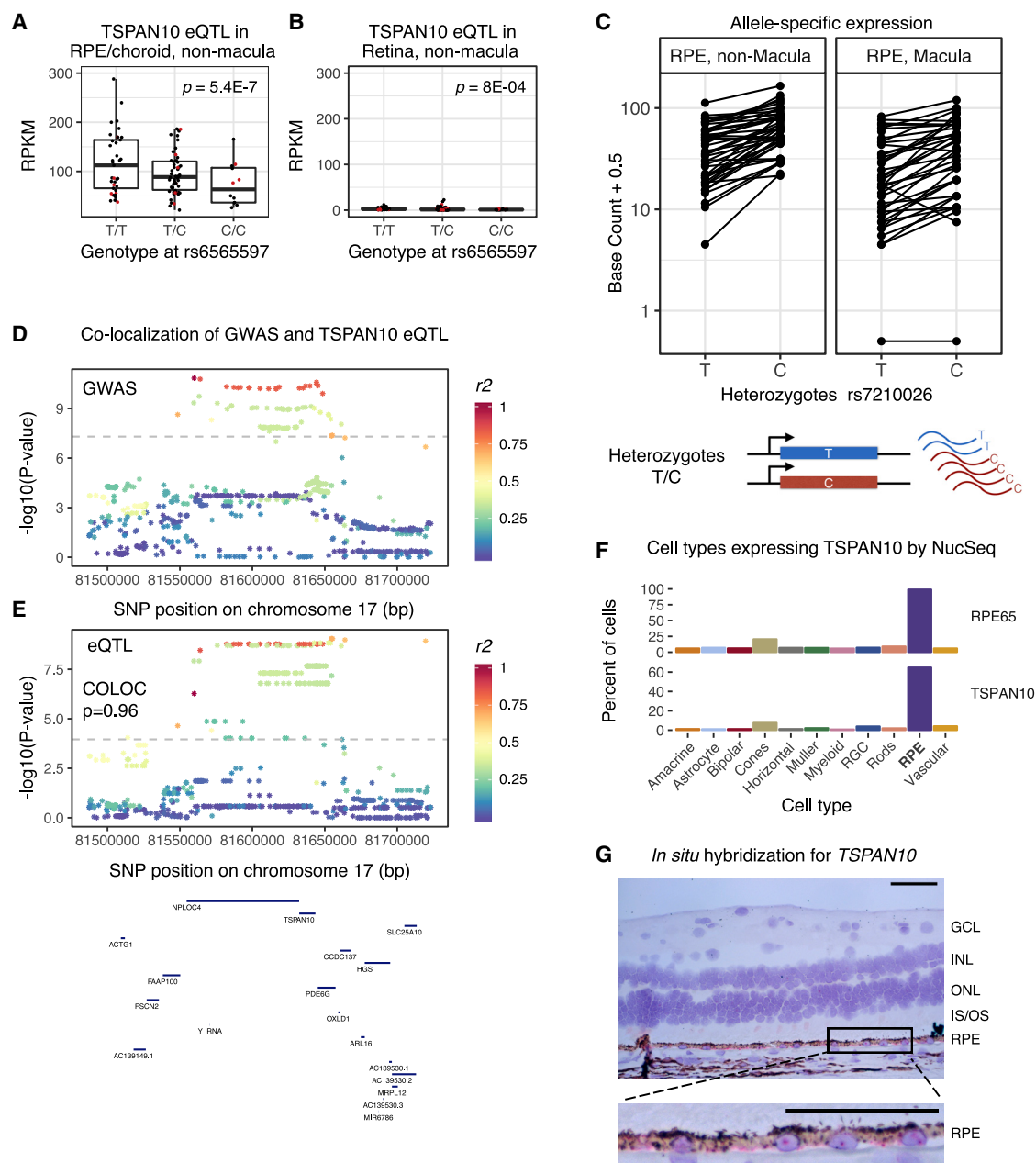


Figure 5. TSPAN10 eQTL in RPE/Choroid Co-localizes with GWAS Locus for Risk of AMD

(A and B) Boxplots summarizing eQTL for *TSPAN10* in RPE/choroid (A) or retina (B), where each dot is a donor, control donors are shown as black dots and AMD donors as red dots, expression as normalized RPKM (reads per kilobase of transcript per million mapped reads) is on the y axis, and donors are grouped by their genotype at the peak SNP reported in the AMD GWAS at this locus on the x axis. The p values indicate the eQTL results for regression of expression levels against the SNP. Boxplots are drawn from 25th to 75th percentiles, horizontal bars are medians, and whiskers show ranges. Donor n = 107 for RPE/choroid and n = 111 for retina tissues.

(C) Allele-specific expression at a coding SNP of *TSPAN10* is shown in heterozygotes. For each donor, counts for *TSPAN10* reads (y axis) are separated by strand on the basis of the genotype of the read at this SNP (x axis). Read counts from each strand are connected by a line by donor. Donor n = 39 for RPE macula and n = 46 for RPE non-macula.

(D and E) Co-localization of genetic association signals for risk of AMD (D) and *cis*-eQTL of *TSPAN10* in RPE/choroid tissue (E). The color of each point represents the Pearson's r^2 value relative to the peak SNP in the GWAS for AMD. Gene models at the locus are shown in the bottom panel.

(F) NucSeq of human eyes shows that expression of *TSPAN10* is observed mainly in RPE cells. Expression of cell type marker *RPE65* is restricted to RPE cells. (G) *In situ* hybridization of *TSPAN10* in retina and RPE/choroid shows expression of *TSPAN10* in RPE cells (red) in a representative image from n = 2 donors. GCL, ganglion cell layer; INL, inner nuclear layer; ONL, outer nuclear layer; IS, inner photoreceptor segments; OS, outer photoreceptor segments; RPE, retinal pigment epithelium. Scale bar, 50 μ m.

See also Figure S5.

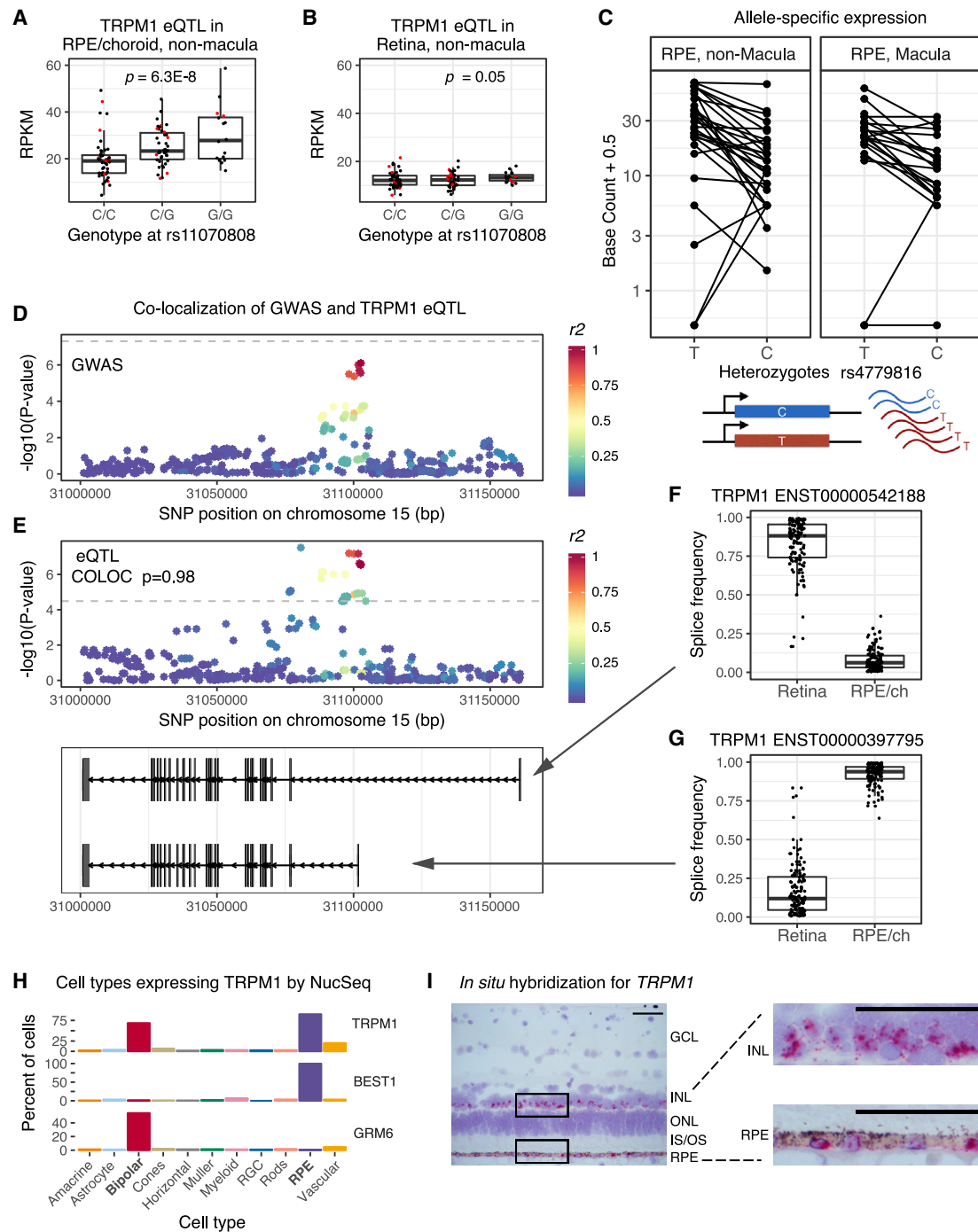


Figure 6. TRPM1 eQTL in RPE/Choroid Co-localizes with GWAS Locus for Risk of AMD

(A and B) Boxplots summarizing eQTL for *TRPM1* in RPE/choroid (A) or retina (B), where each dot is a donor, control donors are shown as black dots and AMD donors as red dots, expression as normalized RPKM is on the y axis, and donors are grouped on the x axis by their genotype at the peak SNP reported in the AMD GWAS at this locus. The p values indicate the eQTL results for regression of expression levels against the SNP. Boxplots are drawn from 25th to 75th percentiles, horizontal bars are medians, and whiskers show ranges. Donor n = 107 for RPE/choroid and n = 111 for retina tissues.

(C) Allele-specific expression at a coding SNP of *TRPM1* is shown in heterozygotes. For each donor, counts for *TRPM1* reads (y axis) are separated by strand on the basis of the genotype of the read at this SNP (x axis). Read counts from each strand are connected by a line by donor. Donor n = 21 for RPE macula and n = 29 for RPE non-macula.

(D and E) Co-localization of genetic association signals for risk of AMD (D) and *cis*-eQTL of *TRPM1* in RPE/choroid tissue (E). The color of each point represents the Pearson's r^2 value relative to the peak SNP in the GWAS for AMD. Gene models at the locus are shown in the bottom panel.

(legend continued on next page)

DISCUSSION

The interplay of genetic and environmental factors is central to the risk of AMD and the progression of the disease from early to advanced stages (Fritsche et al., 2016; Yan et al., 2018). Leveraging our knowledge of the strong genetic component underlying AMD has improved our understanding of disease pathophysiology and shaped therapeutic strategies (Handa et al., 2019; Holz et al., 2014). GWAS provide a unique opportunity for elucidating the mechanisms underlying disease risk, and co-localization between GWAS signals and eQTL for genes in these loci facilitates identification of genes associated with disease risk. Because the macula is the primary region affected in AMD, the absence of a macula in most pre-clinical *in vivo* models limits our efforts to understand disease mechanisms underlying AMD.

Here we present a wide range of datasets from human eye tissues, including retinal and RPE/choroid transcriptomes in peripheral and macular regions, at the single-cell level and in bulk tissue. We observed large differences in expression between retina and RPE/choroid samples dissected from the macula compared with the periphery (Figures 1 and S1; Data S1). Differential expression patterns between macular and peripheral retina in our dataset were consistent with a recent study comparing the single-cell transcriptomes of macaque monkeys (Peng et al., 2019). Similarly, the comparison of macular and peripheral regions in the RPE/choroid were also consistent with previous studies with smaller sample sizes (Li et al., 2014; Menon et al., 2019; Voigt et al., 2019; Whitmore et al., 2014). Our eQTL analysis identified genes whose expression is under genetic control in the macula regions. We found many genes with *cis*-eQTL only in macula samples from the RPE/choroid (3,048) compared with non-macula, or in the macular retina (4,122) compared with non-macula retina. These results are particularly relevant because AMD lesions develop in the macula.

We identified 15 putative causal genes for AMD, including *TSPAN10*, *TRPM1*, *HTRA1*, and *RDH5*. Consistent with a previous study of eQTL in the retina (Ratnapriya et al., 2019), we also found causal evidence for *PILRB*/*PIRLA*, *B3GLCT*, *BLOC1S1*, and *TMEM199*. The *BLOC1S1*-*RDH5* locus appears to be a complex locus with two potential causal genes, where *BLOC1S1* is an eQTL in the retina, and *RDH5* is an eQTL in the RPE. Co-localization between GWAS signals for AMD and eQTL for *PILRB* and *PIRLA* suggests that one or both of these genes drive risk of AMD, but strong LD in the region prevents us from further narrowing down the candidate genes in the locus. It is important to note that AMD risk genes need not be expressed in the eye and that genetic effects do not necessarily manifest at the RNA expression level. A recent study of the human plasma proteome identified protein QTL (pQTL) (Sun et al., 2018), for several AMD

candidate genes, including *APOE*, *C9*, *CFB*, *CFH*, *CFI*, *MMP9*, *PRLR*, *VTN*, and *VEGFA*. Consistent with previous hypotheses, *APOE*, *CFB*, and *CFH* are postulated as causal genes for AMD on the basis of co-localization between genetic association with AMD and plasma pQTL for these genes. It is also possible that distinct eQTL may surface specifically in the disease state, as many environment-specific eQTL have been observed in both humans (Romanoski et al., 2010) and mice (Orozco et al., 2012). Although we did not have a sufficient number of AMD donors in the present study to identify AMD-specific eQTL, it would be of great interest to compare eQTL from control and AMD donors in a large enough cohort in the near future.

Expression of AMD candidate genes was enriched in RPE nuclei, with the largest number of genes (18 of 47; 38%) expressed in RPE relative to other cell types (Figure 4). Notably, 6 of the 15 putative causal genes exhibited *cis*-eQTL specifically in the RPE/choroid, which we could not have uncovered using eQTL from the retina or other publicly available tissue eQTL. Of these 6 genes, *TRPM1* was expressed in both RPE and BCs. Mutations in *TRPM1* have been associated with night blindness in humans, mice, and horses (Miraldi Utz et al., 2018; Peachey et al., 2012; Sandmeyer et al., 2012). These mutations have been implicated in electrophysiological changes in BC activity (Peachey et al., 2012). In our study, we found co-localization of the GWAS signals for AMD and the eQTL only in the RPE/choroid and not in the retina. We found evidence for differential transcription start site use between RPE and retina samples, with the RPE-preferred start site overlapping with the eQTL and GWAS risk peaks. Binding of MITF to the promoter of this start site has been observed in melanocytes (Zhiqi et al., 2004), while polymorphisms in MITF have been shown to reduce *TRPM1* expression in RPE in developing mice (Miller et al., 2004). Taken together, these results suggest that the genetic risk of AMD at the *TRPM1* locus is driven by differential regulation of *TRPM1* in RPE, and not in BCs.

Genetic variation driving decreased expression of *TSPAN10* in the RPE is also responsible for increased risk of AMD. *TSPAN10* appears selectively expressed in melanin-producing cells such as melanocytes and RPE cells. Consistent with this expression pattern, polymorphisms in *TSPAN10* have been associated with response to UV light in fish (Lu et al., 2015), hair color (Hysi et al., 2018; Morgan et al., 2018), and eye phenotypes such as strabismus (Plotnikov et al., 2019), macular thickness (Gao et al., 2019), astigmatism (Shah et al., 2018), and eye color (Liu et al., 2010) in humans. Members of the TspanC8 tetraspanin family have been shown to interact with and regulate the intracellular trafficking of the extracellular metalloprotease ADAM10 (Haining et al., 2012). Co-expression analyses in our bulk RNA-seq data show that *TSPAN10* is highly correlated with genes involved in ion transport and vesicle

(F) Bulk RNA-seq shows that *TRPM1* splice variant ENST00000542188 is expressed in the retina (n = 124 donors).

(G) Bulk RNA-seq shows that *TRPM1* splice variant ENST00000397795 is expressed in the RPE/choroid (n = 125 donors).

(H) NucSeq of human eyes shows that expression of *TRPM1* is observed in RPE and ON bipolar cells. *BEST1* is a marker for RPE cells, and *GRM6* is a marker for ON bipolar cells.

(I) *In situ* hybridization of *TRPM1* in retina and RPE/choroid shows expression of *TRPM1* in RPE cells (red), and in the inner nuclear layer where bipolar cells are located (red), in a representative image from n = 3 donors. GCL, ganglion cell layer; INL, inner nuclear layer; ONL, outer nuclear layer; IS, inner photoreceptor segments; OS, outer photoreceptor segments; RPE, retinal pigment epithelium. Scale bar, 50 μ m.

trafficking, suggesting a potential function for *TSPAN10*. Because vesicle trafficking plays a crucial role in melanin synthesis and recycling of photoreceptors by the RPE, this hypothesized function for *TSPAN10* could explain its role in susceptibility to risk of AMD. Further research on the function of *TSPAN10* in RPE biology is needed to understand the role of *TSPAN10* in AMD disease etiology.

We believe the rich datasets generated in this study are a valuable resource for the ocular research community and will be especially useful for the study of retinal degenerative diseases. Because the majority of the tissue analyzed came from control donors, our datasets can also be used to explore basic human eye biology, and disease associations beyond AMD. Our NucSeq results offer subtype-specific marker genes for aged human retina and RPE and lay the groundwork for future comparative studies of control and diseased eyes at the single-cell level. These transcriptomics datasets are complemented with eQTL in eye tissues including RPE/choroid and retina in macula and non-macular regions, to provide a framework for understanding the genetic and biological mechanisms underlying risk for ocular diseases. Our results highlight the importance of examining the tissues and cell types relevant to the disease biology. Future work examining pQTL in the eye, and single-cell transcriptomics in AMD patients will help us identify additional causal genes and elucidate the molecular pathways underlying AMD.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- **KEY RESOURCES TABLE**
- **LEAD CONTACT AND MATERIALS AVAILABILITY**
- **EXPERIMENTAL MODELS AND SUBJECT DETAILS**
 - Human donor eyes
 - Donor eyes for bulk RNA-seq
 - Donor eyes for NucSeq
 - Donor eyes for *in situ* hybridization
- **METHOD DETAILS**
 - Dissection and RNA extraction for bulk RNA-seq
 - RNA-seq
 - Differential expression in bulk RNA-seq
 - Alternative splicing detection
 - SNP Genotyping, QC and imputation
 - cis-eQTL analysis
 - Allelic imbalance
 - Cis-trans test
 - Co-localization of genetic associations
 - Single nuclei RNA-seq (NucSeq) with 10X Chromium
 - Analysis of single-nuclei RNA-seq data
- **QUANTIFICATION AND STATISTICAL ANALYSIS**
- **DATA AND CODE AVAILABILITY**
 - Data Resources

SUPPLEMENTAL INFORMATION

Supplemental Information can be found online at <https://doi.org/10.1016/j.celrep.2019.12.082>.

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AUTHOR CONTRIBUTIONS

L.D.O., M.v.L.C., J.A.H., H.-H.C., and M.J.T. conceived and designed the study. L.D.O., H.-H.C., M.v.L.C., M.J.T., and J.A.H. wrote the manuscript. L.D.O. and J.A.H. analyzed the data. H.-H.C. performed the NucSeq experiments. H.-H.C., S.S.N., and P.C. performed and interpreted the ISH experiments. R.A. and C.E. performed ocular tissue dissection. C. Cox and K.J.K. performed RNA and DNA extraction and quantification from bulk tissues. Z.M. and Y.-J.C. performed the RNA-seq experiments. L.D.G. built a pipeline for NucSeq alignment. T.B. built a pipeline for SNP imputation. C. Clarke built the eQTL website. A.D. and B.Y. performed genotype QC and provided guidance on genetic analyses.

DECLARATION OF INTERESTS

L.D.O., H.-H.C., C. Cox, K.J.K., R.A., C.E., P.C., S.S.N., Y.-J.C., Z.M., A.D., L.D.G., C. Clarke, T.B., B.Y., M.J., M.J.T., M.v.L.C., and J.A.H. were employees of Genentech at the time these studies were carried out.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Biological samples		
Human eye tissue, control and AMD	Lion's Eye Institute for Transplant and Research (Tampa, FL)	https://lionseyeinstitute.org/ocular-research-center/
Chemicals, Peptides and Recombinant Proteins		
cOmplete™ mini protease inhibitor cocktail	Roche	CAT# 11836153001
Ambion™ RNase inhibitor, cloned, 40U/μl	Life Technologies	CAT# AM2682
SUPERase-In™ RNase inhibitor (20U/μl)	Life Technologies	CAT# AM2696
UltraPure™ BSA (50 mg/ml)	Life Technologies	CAT# AM2616
DAPI (1mg/ml)	Thermo-Fisher	CAT# D1306
DTT, molecular grade, 100 μM	Promega	CAT# P1171
Hematoxylin Novocastra counterstain	VWR	CAT# 103255-766
CytoSeal-60	Thermo Scientific	CAT# 23244257
Critical Commercial Assays		
Chromium single cell 3' gene expression kit v3	10X Genomics	CAT# 1000075
Chromium chip B single cell kit	10X Genomics	CAT# 1000073
Chromium i7 multiplex kit	10X Genomics	CAT# 120262
Chromium single cell 3' gene expression kit v2	10X Genomics	CAT# 120237
Chromium chip A single cell kit	10X Genomics	CAT# 1000009
Agilent RNA 6000 Nano kit	Agilent	CAT# 5067-1511
Agilent High Sensitivity DNA kit	Agilent	CAT# 5067-4626
Fragment Analyzer HS RNA kit	Agilent	CAT# DNF-472-0500
High Sensitivity D1000 ScreenTapes	Agilent	CAT# 5067-5584
High Sensitivity D1000 reagents	Agilent	CAT# 5067-5585
KAPA Library Quant	KAPA Biosystems	CAT# KK4824
RNeasy Mini kit	QIAGEN	CAT# 74106
DNeasy Mini kit	QIAGEN	CAT# 69506
Human Omni 2.5-8v1 SNP BeadChip array	Illumina	CAT# 20037363
TruSeq RNA Sample Preparation Kit v2	Illumina	CAT# RS-122-2001
RNAscope 2.5 HD Reagent Kit-RED	Advanced Cell Diagnostics	CAT# 322350
RNAscope probe Hs-TRPM1, C1 (custom order)	Advanced Cell Diagnostics	CAT# 816061
RNAscope probe Hs-TSPAN10, C1	Advanced Cell Diagnostics	CAT# 483281
Deposited Data		
Raw data files for retina and RPE/choroid RNA-seq	This paper	GEO: GSE135092
Raw data files for NucSeq	This paper	GEO: GSE135133
NucSeq results visualization	This paper	Broad Institute Single Cell Portal: SCP484
Eye eQTL searchable database	This paper	eye-eqtl.com
Software and Algorithms		
GSNAP	Wu and Nacu, 2010	http://research-pub.gene.com/gmap/
R Package HTSeqGenie: A NGS analysis pipeline	Pau and Reeder, 2019	https://doi.org/10.18129/B9.bioc.HTSeqGenie
R package DESeq2	Love et al., 2014	https://doi.org/10.18129/B9.bioc.DESeq2
R package Limma	Law et al., 2014	https://doi.org/10.18129/B9.bioc.limma
R package SGSeq	Goldstein et al., 2016	https://doi.org/10.18129/B9.bioc.SGSeq
R package MatrixEQTL	Shabalin, 2012	https://cran.r-project.org/web/packages/MatrixEQTL/index.html

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
R package PEER	Stegle et al., 2012	https://github.com/PMBio/peer
R package COLOC	Giambartolomei et al., 2014	https://cran.r-project.org/web/packages/coloc/index.html
Samtools	Li et al., 2009	http://samtools.sourceforge.net/
Beagle	Browning and Browning, 2016	http://faculty.washington.edu/browning/beagle/beagle.html#download
LiftOver	Hinrichs et al., 2006	https://github.com/konstantint/pyliftover
PLINK	Chang et al., 2015	https://github.com/chrchang/plink-ng
Seurat version 3	Butler et al., 2018	https://cran.r-project.org/web/packages/Seurat/index.html
Other		
Human reference genome NCBI build 38, GRCh37	Genome Reference Consortium	https://www.ncbi.nlm.nih.gov/projects/genome/assembly/grc/human/
GENCODE Human v27	GENCODE	https://www.encodegenes.org/human/release_27.html

LEAD CONTACT AND MATERIALS AVAILABILITY

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Jason A. Hackney (jasonah@gene.com). This study did not generate new unique reagents.

EXPERIMENTAL MODELS AND SUBJECT DETAILS

Human donor eyes

Post-mortem human eyes from 129 human donors for bulk RNA-seq, 5 donors for single-nucleus RNA-seq and 3 donors for *in situ* hybridization were procured by the Florida Lions Eye Institute for Transplant and Research (Tampa, FL). Clinical records and a family questionnaire were obtained for all donors. The studies using human tissues were performed in accordance with FDA regulations and the Eye Bank Association of America (EBAA) medical standards regarding utilization of human tissue. Written, informed consent was obtained from all donors who provided human samples in accordance with the guidelines of the Declaration of Helsinki for research involving the use of human tissue. The protocols for these studies were approved by the Pharma Repository Governance Committee at Genentech that serves as the Genentech/Roche Institutional Ethical Committee to ensure that research on human samples stored in Genentech bio-repositories is performed in accordance with the subject's informed consent and with global ethical guidelines.

Donor eyes for bulk RNA-seq

Donors profiled by bulk RNA-seq were 53 females and 76 males, ranging from 59 to 98 years of age. From these, 106 donors had no history of ocular diseases and 23 donors were previously diagnosed with AMD. 5X low-power images were obtained from the posterior segment of the donor eyes using a Leica M165C stereo microscope. Exclusion criteria for donors included a history of diabetes or glaucoma. Donor eyes were excluded from this study if, upon examination of donor macular images, there were signs of retinal pathology indicative of ocular disease not related to AMD or if the retina and/or RPE cell layers were damaged due to dissection or transport of the eyes.

Both eyes from each donor were enucleated 4 h post-mortem or less and preserved in RNAlater immediately after collection. Right eyes were further processed as replacement only if the RNA quality did not meet the required RNA integrity (RIN < 8). Grading was performed after removal of the anterior segment, vitreous, and retina and performed according to the Minnesota Grading System for post-mortem human eyes, based on the Age-Related Eye Disease Study grading system (Olsen and Feng, 2004). Only donor eyes with category 1 disease (controls) and category 4 disease (advanced AMD) were used for this study. Donors previously diagnosed with AMD included any advanced AMD, with no specification of wet or dry form of the disease. Eyes from donors in which ophthalmic history did not match the macroscopic observation in either or both eyes from a single donor were not further processed.

Donor eyes for NucSeq

Donor eyes with no retinal disease diagnosis were enucleated and flash frozen in LN2 within 5 h post-mortem, shipped on dry ice and stored at -80°C . 108955 (right eye) is from a 78 year old female donor, post-mortem interval (PMI) = 3:40 h, RNA integrity number (RIN) = 8.0; 120974 (left eye) is from a 79 year old male donor, PMI = 2:40 h, RIN = 8.0; 109829 (left eye) is from a 77 year old female donor, PMI = 4:08 h, RIN = 8.4; 109373 (right eye) is from a 70 year old male donor, PMI = 4:40 h RIN = 8.6; 110814 (right eye) is from a 65 year old male donor, PMI = 4:25 h, RIN = 8.5.

Donor eyes for *in situ* hybridization

Control donor eyes with PMI <6 h were formalin-fixed by injection into the globe and immersed for 24–48 h and transferred into ethanol for shipping at ambient temperature. HP135097 is from a 73 year old female donor, HP135100 is from a 88 year old male donor and HP135102 is from a 79 year old female donor.

METHOD DETAILS

Dissection and RNA extraction for bulk RNA-seq

The macula is fully contained within the boundaries of superior and inferior temporal vascular arcades and is easily visualized. After the macula was dissected out from the peripheral fundus using dissecting scissors, the macular retina was separated from the RPE and choroid underneath the retina. Total RNA was isolated from tissues using RNeasy Mini kit (QIAGEN) including the on-column DNase digestion.

RNA-seq

Quality control of samples was done to determine RNA quantity and quality prior to processing for RNA-seq. The concentration of RNA samples was determined using NanoDrop 8000 (Thermo Scientific) and the integrity of RNA was determined by Fragment Analyzer (Advanced Analytical Technologies). Samples with RNA integrity number (RIN) below 8 were excluded from further analysis. 0.5 ug of total RNA was used as an input material for library preparation using TruSeq RNA Sample Preparation Kit v2 (Illumina). Size of the libraries was confirmed using 2200 TapeStation and High Sensitivity D1000 screen tape (Agilent Technologies) and their concentration was determined by qPCR-based method using Library quantification kit (KAPA). The libraries were multiplexed and then sequenced on Illumina HiSeq2500 (Illumina) to generate 30 million, single-end, 50 base pair reads. Sequencing data analysis was performed as previously described (Durinck et al., 2015). Briefly, sequencing reads were mapped to the reference human genome (GRCh38), using the GSNAP short read aligner (Wu and Nacu, 2010). Transcript models used for differential expression were based on GENCODE Basic annotations v. 27. Expression was quantified using HTSeqGenie (Pau and Reeder, 2019) as reads per kilobase of gene model per million total reads (RPKM), and normalized using size factors (nRPKM) computed using DESeq2 (Love et al., 2014).

Differential expression in bulk RNA-seq

We tested for differential expression between bulk RNA-seq samples using linear modeling with the voom/limma package (Law et al., 2014), and adjusted p values for multiple testing using the Benjamini-Hochberg method. Genes were considered differentially expressed if they had adjusted p value < 0.05 and absolute fold change > 2.

Alternative splicing detection

We used the SGSeq package (Goldstein et al., 2016) to quantify use of splice events in the bulk RNA-seq data. We used transcripts that are part of GENCODE Basic v. 27 annotations to identify splice features to count. To determine alternative use of splice events, we used a beta-binomial model to fit the proportions of a particular splice variant at a given splice event. The model was run separately on macular and peripheral tissue, comparing RPE samples to retinal samples, and incorporated covariates for technical batching of the RNA-seq samples.

SNP Genotyping, QC and imputation

DNA was extracted from conjunctiva tissues using QIAGEN DNeasy, and genotyped using the Illumina 2.5M OMNI SNP array. Samples were excluded if the call rate was less than 90%. Samples were removed if they showed excess heterozygosity with more than three standard deviations of the mean. Identity by descent analysis was used to detect and filter out relatedness in the dataset. For identity by descent analysis (patient relatedness), the PI_HAT cutoff for pairwise relatedness was 0.4 or higher. Principal components were generated including the HapMap populations, and samples were excluded if they did not cluster near the reference European population (CEU). SNPs were filtered out if they had a MAF less than 5%, missingness was higher than 5%, or if the significance level for the Hardy-Weinberg equilibrium was less than 1×10^{-8} . These criteria left 1,890,601 SNPs after QC. We used these SNPs for imputation using Beagle (Browning and Browning, 2016) and the 1000 Genomes project Phase3. Imputed SNPs were filtered out again after imputation for Hardy-Weinberg Equilibrium ($p < 1 \times 10^{-8}$), missingness in more than 10% of the samples, or MAF less than 5%. SNP positions were mapped to GRCh38 using LiftOver (Hinrichs et al., 2006) for a set of 10,125,014 SNPs. To avoid outlier effects, SNPs were excluded if the smallest genotype group had less than 3 individuals, for a final set of 5,671,011 SNPs. QC and filtering steps were performed using PLINK (Chang et al., 2015).

cis-eQTL analysis

We used the R package MatrixEQTL (Shabalov, 2012) to identify SNPs whose genotypes were associated with gene expression of nearby genes, also known as *local* or *cis*-eQTL. For every gene, we restricted eQTL discovery to SNPs located within 1Mb upstream and 1Mb downstream of the gene. For eQTL analysis, we selected “expressed genes” with normalized RPKM (nRPKM) greater than 0.1 in at least 10 donors, resulting in 31,560 genes. We transformed expression levels using rank based inverse normal transformation. We computed PEER factors from the transformed expression with the R package PEER (Stegle et al., 2012), with the aim to

remove effects of confounding variables from the gene expression matrix. Our choice of 15 PEER factors is a conservative approach that aims to capture even moderate-to-low strength confounders. We used the following linear regression to identify SNPs associated with gene expression while controlling for covariates, including 3 ancestry PCs, sex as determined from RNA-seq, RNA-seq batch, genotyping batch, AMD status, and 15 PEER factors:

$$\text{Expression} \sim \text{SNP} + \text{ancestry} + \text{sex} + \text{RNAseq batch} + \text{Genotype batch} + \text{PEER factors}$$

After genotyping and QC, 98 control and 23 AMD donors were retained ($n = 121$) for subsequent eQTL analysis. The number of donors with both genotyping and expression data per tissue was 106 for retina macula, 111 for retina non-macula, 106 for RPE macula, and 107 for RPE non-macula.

Allelic imbalance

For each SNP identified in exonic and UTR regions of genome, we counted the number of reference and alternate bases using the samtools (Li et al., 2009) pileup tool. Only bases with quality >23 were included in the count. Positions with an average sequencing depth ≥ 1 , with at least 5 heterozygous individuals were included in the analysis. Each tissue was analyzed separately. To test for allelic imbalance, we fit a beta binomial model, using a maximum likelihood estimator, to the reference and alternate base counts at each position for all individuals called heterozygous in the imputed genotypes, including a covariate to account for RNA-seq batching. Because imputation using Beagle codes alleles to major and minor allele, we recoded our reference allele to be the major allele observed in the RNA-seq cohort. Thus, our estimates of allele-specific expression are consistent in direction with the eQTL effect estimated by MatrixEQTL. Since each SNP is tested independently, and we do not make any assumptions about LD structure, we treat each SNP as a self-contained test. For each SNP, we estimate a p value for allele specific expression. We use the false discovery rate to account for multiple testing (Benjamini and Hochberg, 1995). Since we only tested for allele-specific expression in genes with sufficient RNA-seq data at exonic SNPs, it is likely that we did not observe additional allele-specific expression for many eGenes in part due to lack of genes with sequencing coverage in reads that were both exonic and polymorphic.

Cis-trans test

The test for allele specific expression described above makes use of all individuals heterozygous for a SNP. This test will only identify allele specific expression if the measured SNP is in high LD with a *cis*-eQTL. If a non-coding *cis*-eQTL SNP is heterozygous in only a subset of the total pool of heterozygotes at a coding SNP, the coding SNP will likely not show allele specific expression. However, if we restrict our allele-specific expression analysis to include only individuals that are heterozygous for both the *cis*-eQTL SNP and the coding SNP, we expect to find allele-specific expression in that subset of individuals. We can test for this using the same beta-binomial model as before, on the allele counts from the samples that are heterozygous for the exonic SNP and the putative *cis*-eQTL SNP. If we repeat procedure for all exonic SNPs that co-occur with a putative eQTL SNP, we obtain an aggregated value for all SNPs that are heterozygous for the putative eQTL SNP. Since we do not have phased genotypes, we cannot determine which allele at a heterozygous SNP is on the same strand as the eQTL SNP. So, we use Fisher's method for combining p values to determine a per-eGene/SNP meta-p value. We repeat this procedure for all pairs of eQTL SNPs and eGenes that are found to be significant in the standard eQTL analysis.

Co-localization of genetic associations

To find loci where genetic variation is driving both expression and disease risk, we used co-localization between eye eQTL and GWAS genetic signals using the COLOC package (Giambartolomei et al., 2014). Briefly, COLOC performs a statistical analysis between two potentially related phenotypes, to ask if they share common genetic causal variant(s) in a given region. The method utilizes SNP variant association p values, the MAFs of these SNPs, and the sample size, and generates a vector for each phenotype, where each SNP is encoded by 0 or 1 to represent no association and association, respectively. The method then assesses support for all possible configurations, i.e., for each pair of binary vectors, to test the following hypothesis: H0: there is no association with either trait; H1: association with trait 1, not with trait 2; H2: Association with trait 2, not with trait 1; H3: Association with trait 1 and trait 2, two independent SNPs; H4: there is an association with trait 1 and trait 2, one shared SNP. The method then uses Approximate Bayes Factor to compute the posterior probability of each of the five configurations. We used the function *coloc.abf* function, using the p values for eQTL and public GWAS for AMD (Fritsche et al., 2016), the sample sizes, and SNP MAFs, with the default function parameters.

Single nuclei RNA-seq (NucSeq) with 10X Chromium

Frozen eye globes from control donors were cryosectioned at 60 μm starting at the optic nerve head and at a plane perpendicular to the optic nerve. Posterior sections with visible RPE layers were collected in batches of 8 for NucSeq (see below). Additional sections of more anterior eye regions were used to assess overall RNA quality. Nuclei were isolated and processed based on methods in Krishnaswami et al. (2016). Briefly, sections were thawed directly in ice cold homogenization buffer (250 mM sucrose, 25 mM KCl, 5 mM MgCl₂, 10 mM Tris buffer pH 8.0, 1 μM DTT, 1X cOmplete™ protease inhibitors with EDTA (Roche), 0.1% v/v Triton X-100, 0.4U/ μL recombinant RNase Inhibitor (Ambion), 0.2U/ μL SUPERase-In (Ambion), 0.2 $\mu\text{g}/\text{ml}$ DAPI) and dounced 15X with a loose pestle in a glass tissue homogenizer (Wheaton). Nuclei released were washed once in homogenization buffer and sorted for

2n DNA content gated on DAPI (linear scale) on a FACS-ARIA II (BD) with cooled chamber. Sorted single nuclei were collected in 1% BSA/1X PBS with 0.2U/μL RNase Inhibitor, concentrated, resuspended in ~50 μL residual buffer, counted manually with a hemocytometer, and immediately processed with 10X Chromium single cell 3' gene expression kit v2 or v3 (10X Genomics) at ~16000 nuclei per library and 2–3 libraries per region, following manufacturer's protocol.

We performed a pilot experiment with one donor eye using the version 2 kit from 10X, and obtained 28,088 nuclei after QC, with an average of 951 genes detected per nucleus. While this represented a much lower proportion of the total transcriptome compared to scRNA-seq, there was sufficient information in this pilot dataset to resolve all major cell types as well as several neuronal subtypes (Figure S2G). For donor eye 108955, sections 1–8 (region 1, most posterior) and 9–18 (region 2) are separately subjected to nuclei extraction and library construction with the 10X Chromium single cell 3' gene expression kit v2. For the remaining 4 donor eyes, only region 2 was used for NucSeq with the v3 kit.

Analysis of single-nuclei RNA-seq data

Alignment pipeline for 10X data

Single-nuclei RNA-seq data were processed with an in-house analysis pipeline. Briefly, reads were demultiplexed based on perfect matches to expected cell barcodes. Transcript reads were aligned to the human reference genome sequence (GRCh38) using GSNAP (Wu and Nacu, 2010). Transcript counts for a given gene were based on the number of unique UMIs for reads overlapping in antisense orientation. Since we used RNA derived from nuclei, both exonic and intronic reads were considered for downstream analysis. To account for sequencing PCR errors, one mismatch was allowed when collapsing UMI sequences. Only barcodes with more than 10,000 associated reads were deemed to be true cells and kept for analysis.

Normalization

We normalized UMIs using Seurat version 3 (Butler et al., 2018) with the 'LogNormalize' method. We selected variable genes based on dispersion, then used these to compute principal components and tSNE dimensional reductions with perplexity = 50 and theta = 0.5.

Clustering

We generated clusters of transcriptionally related cells corresponding to cell types or cell type subsets by using graph-based clustering based on the Louvain algorithm, implemented in the Seurat function *FindClusters*. The number of principal components were then used to generate sample clusters varied from 3 to 20 depending on the cell type and number of cells used for clustering.

Doublets, empty droplets, and contaminants

We removed doublets and empty droplets by selecting for droplets with total UMI greater than 1,000 but less than 6,000, and number of genes detected greater than 500 but less than 5,000. We also removed droplets with contaminating, or *ambient*, RNA by excluding libraries found in non-Rod clusters that were expressing rod markers RHO, NRL, and PDE6A. These QC thresholds decreased the size of our dataset from 121,779 cells to 100,055 cells. For a small number of 397 cells, these thresholds were not sufficient to remove droplets that were neither empty, nor doublets, but did contain ambient RNA. We manually removed libraries from these 397 droplets based on expression of cell type markers from more than one major cell type.

Cluster markers

We searched for gene expression markers that were more highly expressed in each cluster, relative to all other clusters, using the *FindAllMarkers* function in Seurat, based on the non-parametric Wilcoxon rank-sum test. Cluster markers were only considered if they were expressed in at least 10% of the cells in the cluster, with a minimum difference of 30% in the fraction of cells expressing the marker between two clusters, and a minimum log fold change in expression of 0.25.

Pseudo-bulk and Differential Expression analysis

Pseudo-bulk datasets are derived from the NucSeq dataset by aggregating the cells of each sample of the same cell type. For n donors and m cell types there are $n \times m$ total possible pseudo-bulks, i.e., aggregates of cells of a single cell type from a single donor. If less than 10 cells of a particular type are present in a given donor, they would be discarded. A single "raw count" expression profile was created for each pseudo-bulk by adding the total number of UMIs for each gene across all the cells. Let $n_{i,j}$ be the raw UMI number of gene i in cell j ; let s_j and c_j indicate the donor sample and cell type, respectively, of cell j . The pseudo-bulks are the set of pairs (s, c) of samples s and cell types c , with $(s_j, c_j) = (s, c)$. The pseudo-bulk count matrix B , with rows indexed by genes and columns indexed by pseudo-bulks is defined as:

$$B_{i,sc} = \sum_{j:(s_j, c_j) = (s, c)} n_{ij}$$

This gave a pseudo-bulk count matrix B which was then normalized to a normalizedCount statistic using then estimateSizeFactors function from DESeq2. Differential expression was performed on this data-set using voom-limma methods for bulk RNA-seq as described above. Results from this pseudo-bulk and DE analysis can be found in Data S3.

We also performed an analysis where NucSeq UMI counts were normalized (using DESeq2 size factors as described above) prior to generating the pseudo-bulk matrix, followed by differential expression analysis. The analyses yielded similar results regardless of normalization prior to or after pseudo-bulk estimation, where the Pearson's correlation between the log2(Fold-change) estimated in

the two approaches ranged between $r = 0.94$ and $r = 0.96$. Representative examples are shown in [Figure S3](#). We note that for a subset of the genes, normalization of NucSeq UMIs prior to pseudo-bulk resulted in overestimation of the magnitude in fold-change in expression, for example see [Figures S3B](#) and [S3D](#).

Test of cell proportions

To compare proportion of cell types across donors, we used a Fisher's exact test where the number of successes equaled the number of cells of a given cell type in each donor, and the number of failures was the total number of cells from a given donor minus the number of successes. This test was repeated for all major cell types. We used the Bonferroni method to correct for repeated tests in 11 major cell types.

In situ hybridization

Fixed globes from control donors were sagittally transected into 2 hemispheres, dehydrated through a series of ethanol followed by xylene, and paraffin-embedded and sectioned at 4 μm . RNAscope (Advanced Cell Diagnostics) probes used were: TRPM1 (custom made C1 probe, 20ZZ probe targeting 2-1065 of NM_002420.6) and TSPAN10 (ACD #483281). *In situ* hybridization was performed with the RNAscope 2.5 HD Reagent Kit-RED (ACD #322350). Briefly, sections of peripheral retinal areas from 3 different donors were baked at 60°C for 1 h, deparaffinized, treated with hydrogen peroxide for 10 min and lightly boiled in Target Retrieval Buffer (ACD) for 15 min. Wax barriers were drawn with ImmEdge pen (Vector) and air-dried overnight. The sections are then treated with Protease Plus (ADC) for 30min at 40°C, and followed by probe hybridization for 2 h at 40°C. Signal amplification was performed following the manufacturer's protocols. The slides were counterstained with Novocastra Hematoxylin (Leica), briefly washed in tap water and mounted in CytoSeal-60 (Thermo Scientific). The ISH brightfield images were acquired on a Zeiss Axioskop 40 microscope.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical methods and software packages, as well as exclusion criteria are described in detail in appropriate sections within [STAR Methods](#). Statistical analyses were performed using R. Exact values of numbers of donors/tissue used are describe in Results, Figure Legends and [STAR Methods](#). In all heatmaps, color is the Z score of the expression level scaled by rows.

DATA AND CODE AVAILABILITY

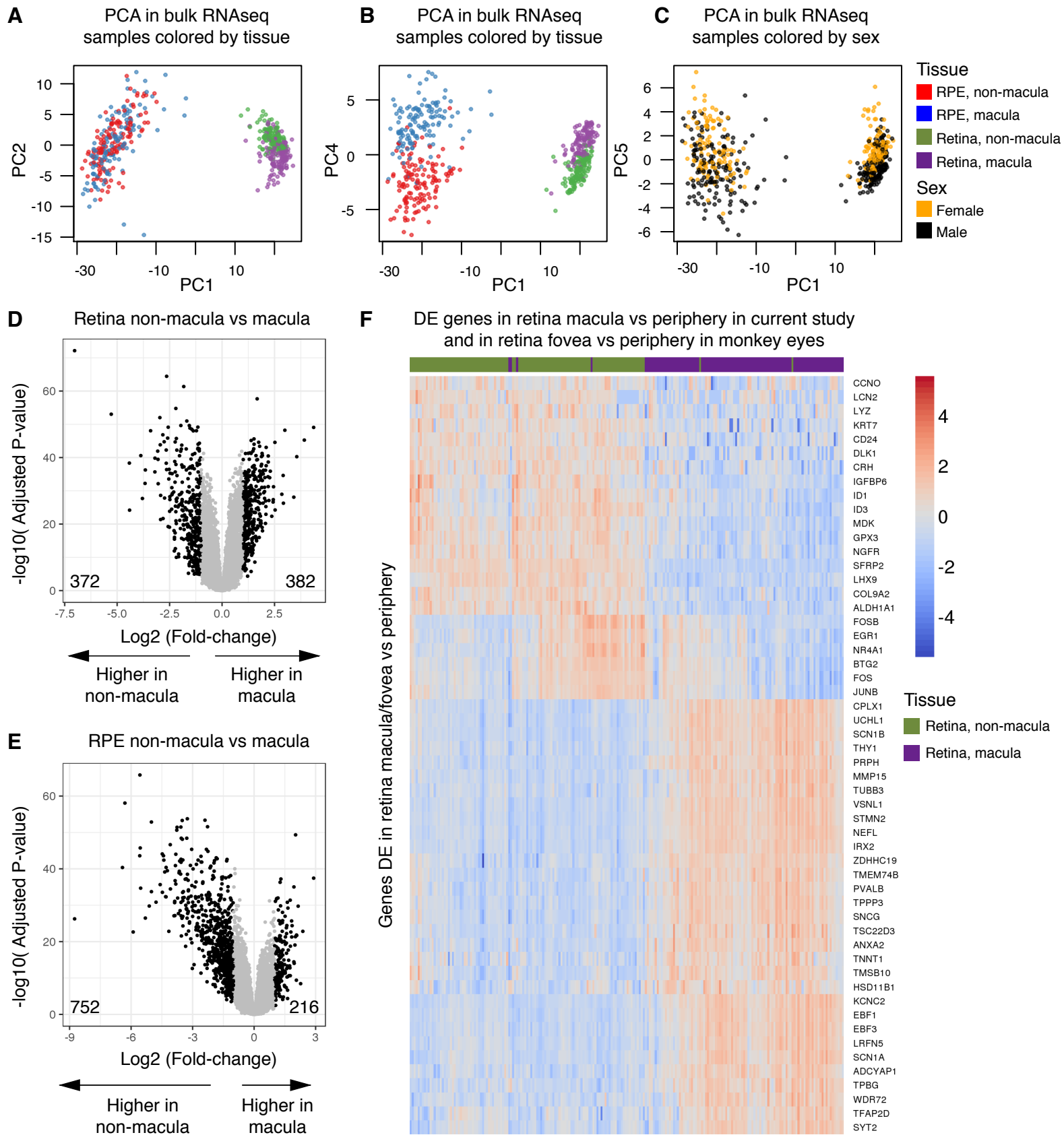
Data Resources

Processed RNAseq data can be found in GEO for bulk RNA-seq from retina and RPE/choroid under GEO: GSE135092, and NucSeq under GEO: GSE135133. NucSeq results can be browsed in the Broad Institute's Single-Cell Portal under study SCP484 at https://singlecell.broadinstitute.org/single_cell. Eye eQTL results can be found at eye-eqtl.com. Further information about sample preparation, data collection, or data processing is described in the method details and can also be directed to the Lead Contact.

Supplemental Information

Integration of eQTL and a Single-Cell Atlas in the Human Eye Identifies Causal Genes for Age-Related Macular Degeneration

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Supplemental Figure S1. Macula and tissue-specific transcriptomes from human eyes. Related to Figure 1.

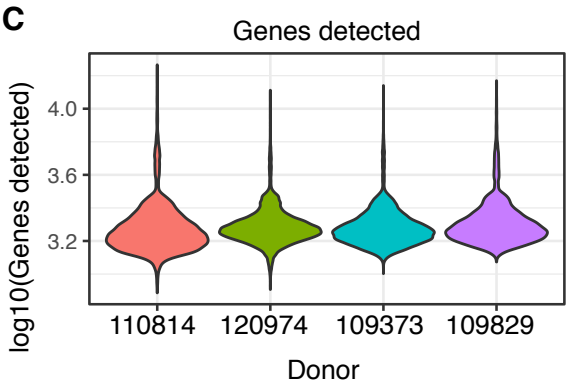
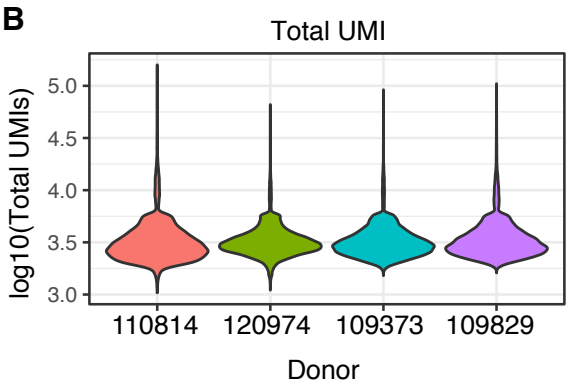
(A)-(C) Principal component analysis, each dot is a sample, and samples are colored based on (A)-(B) tissue, or (C) sex.

(D)-(E) Volcano plots summarizing differential expression analysis in bulk RNAseq, where each dot is a gene, and genes are colored in black if they had an adjusted p-value < 0.05 and fold change > 2. Comparisons are (D) Retina non-macula vs macula, (E) RPE non-macula vs macula.

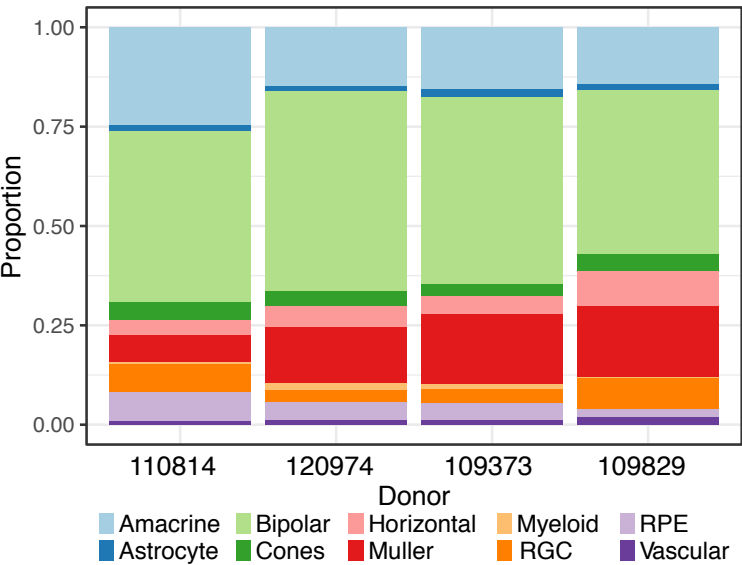
(F) Genes differentially expressed between retina macula and non-macula in our bulk RNAseq samples were consistent with genes reported in monkey fovea versus peripheral retina using scRNAseq. Genes shown in rows are DE in published and in our current datasets with adjusted p-value < 0.05 and fold change > 2. Columns are samples, and the color is the expression level in our dataset represented as a z-score scaled by rows. For all panels, donor n=124 for retina, and n=125 RPE/choroid samples.

A Summary statistics in NucSeq samples

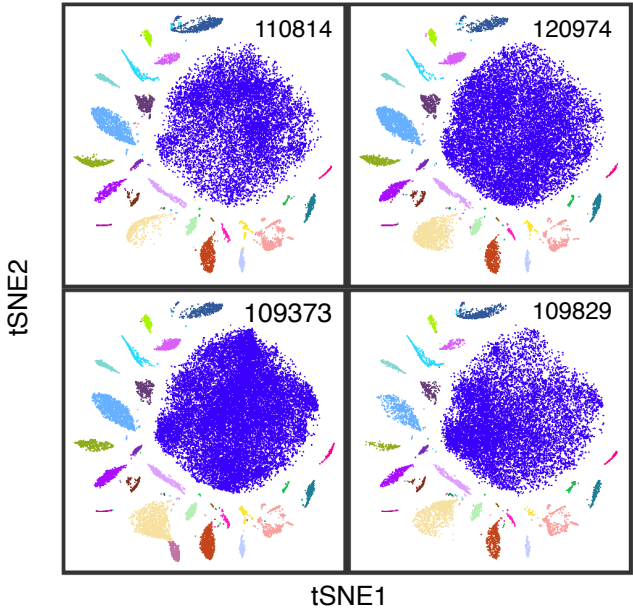
Sample	Donor	Number of nuclei per sample	Mean total UMI	Mean number of genes detected
Sample1	109829	9,619	3,824	2,159
Sample2	109829	9,743	3,791	2,129
Sample3	120974	8,521	3,467	2,018
Sample4	120974	8,601	3,414	1,989
Sample5	120974	9,358	3,339	1,964
Sample6	109373	11,899	3,486	1,990
Sample7	109373	11,815	3,414	1,961
Sample8	109373	11,285	3,387	1,953
Sample9	110814	5,642	3,925	2,060
Sample10	110814	5,715	3,790	2,021
Sample11	110814	7,857	3,711	1,986



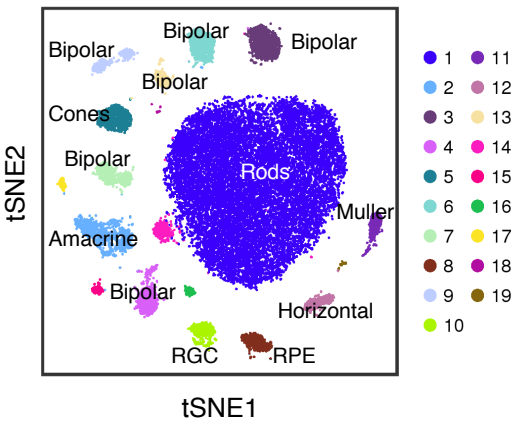
D Proportion of cell types by donor



E Cell type clustering by donor



F NucSeq for all cell types in 10X v2 kit



Supplemental Figure S2. Single-nuclei RNA sequencing in human retina and RPE.
Related to Figure 2.

(A) Summary of sample characteristics for number of nuclei, total UMI and genes detected in NucSeq data by sample.

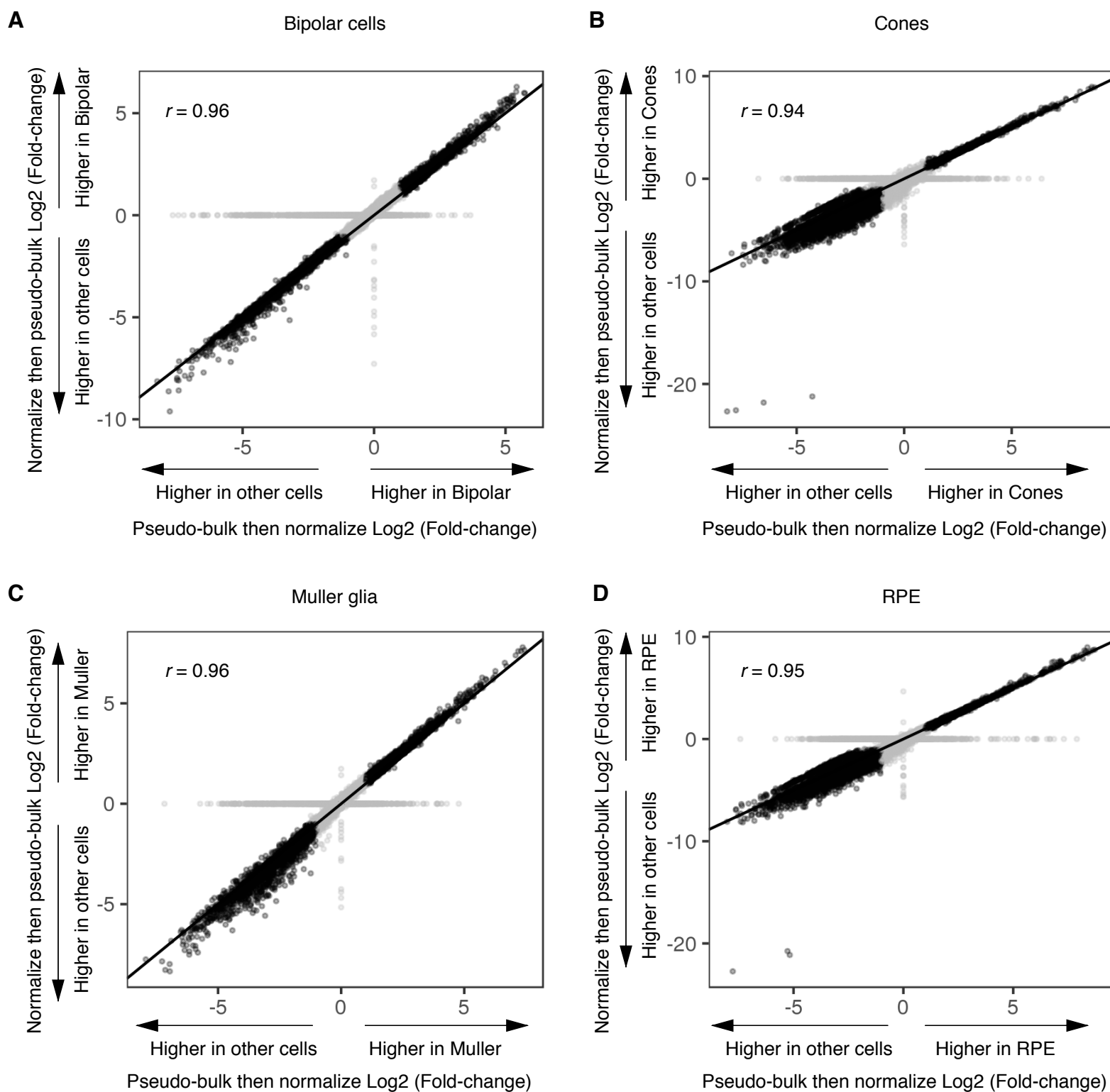
(B) Distribution of total UMIs across all NucSeq donors prepared using the 10X version 3 kit. Violin plots show the distribution of the data and its probability density (donor n=4).

(C) Distribution of the number of genes detected across all NucSeq donors prepared using the 10X version 3 kit. Violin plots show the distribution of the data and its probability density.

(D) Proportion of nuclei assigned to each major cell type across different donors, after excluding rod photoreceptors (donor n=4).

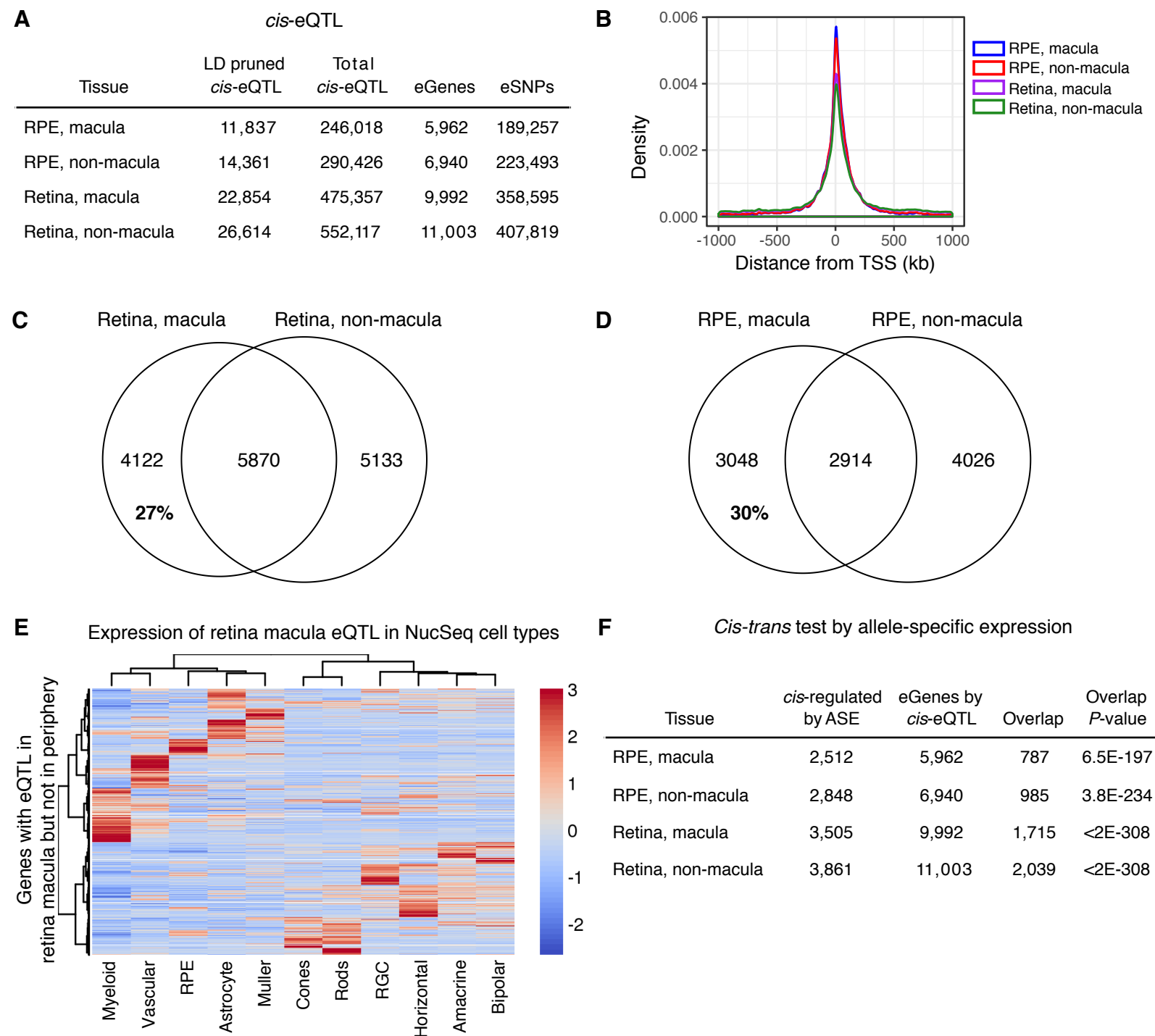
(E) Dimensionality reduction using tSNE of single nuclei RNAseq for all cell types faceted by human donor using the 10X version 3 kit (donor n=4).

(F) Dimensionality reduction using tSNE of single-nuclei RNAseq from a human eye for all cell types identified using the 10X version 2 kit (donor n=1). Each dot is a nucleus. Nuclei are colored by cluster. Cluster labels: rods are 1 and 14; amacrine cells (AC) are 2, 15 and 17; bipolar cells (BC) are 3, 4, 6, 7, 9, 13, 16 and 18; cones are 5; RPE cells are 8; retinal ganglion cells (RGC) are 10; Muller glia and astrocytes are in 11; Horizontal cells are 12; vascular cells are 19.



Supplemental Figure S3. Differential expression analysis examples of NucSeq pseudo-bulk. Related to Figure 3.

(A)-(D) 4-way plots of differential expression analysis using pseudo-bulks for NucSeq data. On the x-axis is DE doing pseudo-bulk followed by normalization, on the y-axis is DE analysis doing normalization in UMI counts prior to pseudo-bulk. The Pearson's r correlation between the two approaches is shown. Each dot is a gene, and genes are colored in black if they had fold change > 2 . Donor $n=4$. Comparisons are shown for (A) bipolar cells, (B) cone photoreceptors, (C) Muller glia, and (D) RPE cells.



Supplemental Figure S4. Summary of eQTL in RPE/choroid and retina. Related to Table 1.

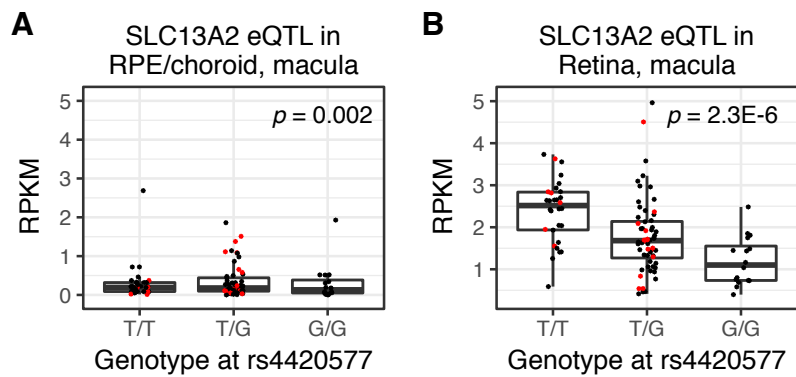
(A) Summary of eQTL by tissue at FDR<5% (donor n=121).

(B) Distribution of the distance between associated SNP and the gene transcription start site, TSS.

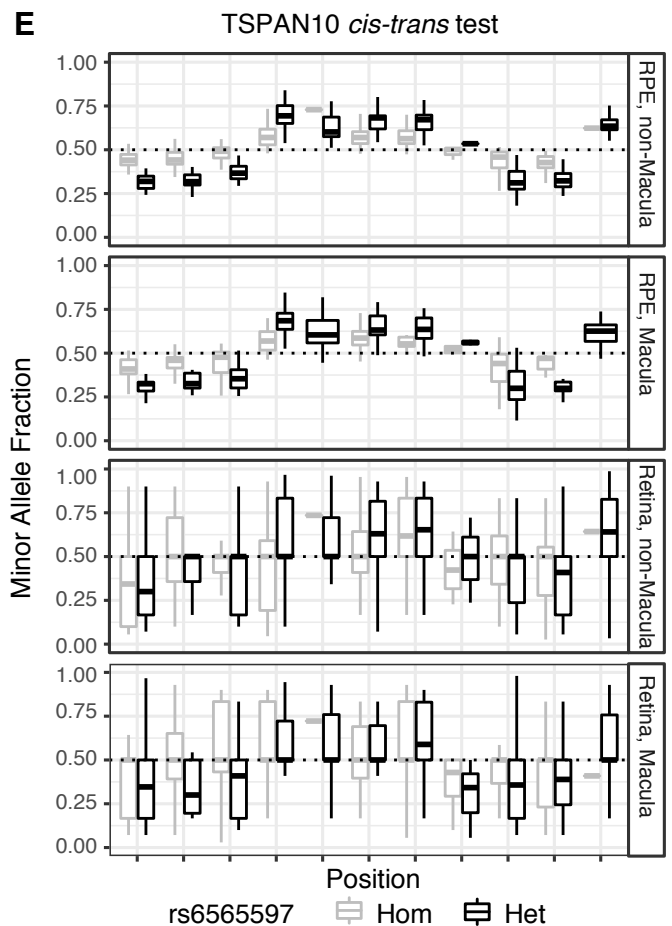
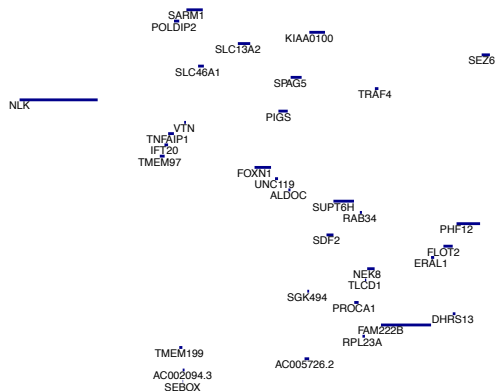
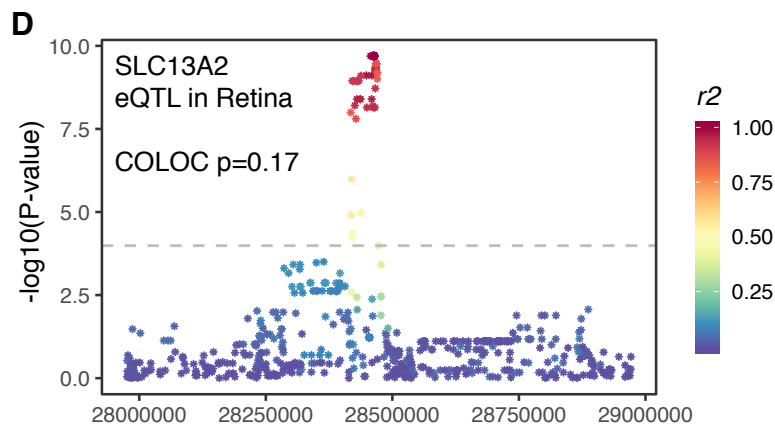
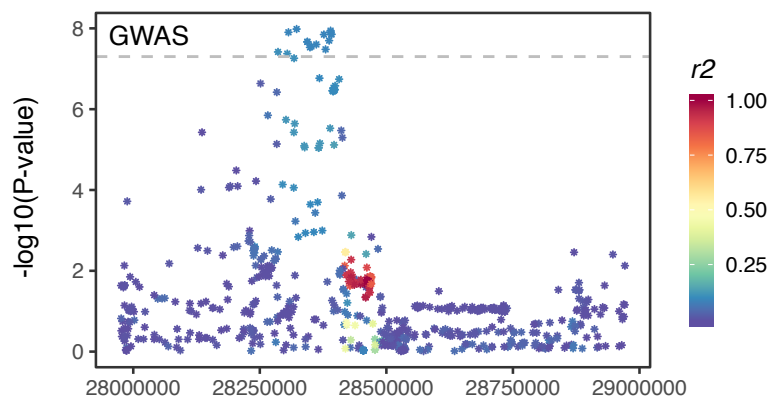
(C)-(D) Venn diagrams showing overlap between *cis*-eQTL identified in macular on non-macular regions in (C) retina and (D) RPE/choroid tissues, at FDR<5%.

(E) Genes that are eQTL in retina macula but not in non-macula (periphery) are shown in rows, the cell types are shown in columns, and the heatmap represents the average expression by cell type in the NucSeq, represented as a z-score scaled by rows (donor n=4).

(F) Overlap between *cis*-eQTL and genes determined to be regulated in *cis* using allele-specific expression (ASE).



C No co-localization of GWAS and SLC13A2 eQTL



Supplemental Figure S5. *Cis*-eQTL and *cis-trans* test. Related to Figure 5.

(A)-(B) Boxplots summarizing eQTL for *SLC13A2* in (A) RPE/choroid and (B) retina. Each dot is a donor, control donors are shown in black dots, and AMD donors in red dots, expression as normalized RPKM is on the y-axis, and donors are grouped by their genotype on the x-axis at the peak SNP reported in the AMD GWAS at this locus. Boxplots are drawn from 25th to 75th percentiles, horizontal bars are medians, and whiskers show ranges. Donor $n=106$ for RPE/choroid and retina tissues.

(C)-(D) Lack of co-localization of genetic association signals for risk of AMD (C) and *cis*-eQTL of *SLC13A2* in retina tissues (D). The color of each point represents the Pearson's r^2 relative to the peak SNP for the eQTL. Gene models at the locus are shown on the bottom panel.

(E) *Cis*-regulation of *TSPAN10* gene expression is observed in RPE/choroid tissues. Boxplots show the distribution across heterozygotes in the ratio of read counts from the minor allele strand over total counts from both strands. Deviation from 0.5 suggests expression is higher from one strand than the other. On the x-axis are multiple SNPs in the locus. At each SNP, donors are grouped by their genotype at the peak SNP reported in the AMD GWAS at this locus. Boxplots are drawn from 25th to 75th percentiles, horizontal bars are medians, and whiskers show ranges.